# CS177 Bioinformatics: Software Development and Applications

## Lecture 16: Epigenomics and Proteomics

Jie Zheng (郑杰)

PhD, Associate Professor

School of Information Science and Technology (SIST), ShanghaiTech University

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# Learning objectives

- At the end of this lecture, you should be able to:
  - Explain biological background of epigenomics
  - Describe ChIP-seq technique and related data analysis
  - Explain basics of ATAC-seq technique
  - Explain basics of Hi-C technique and data
  - Describe biological background of proteomics
  - Explain how mass spectrometry works
  - Define computational problems in proteomics

# Biological Background of Epigenomics

# Epigenetics



- Epigenetics is the science for studying heritable changes of gene expression, without involving changes of the DNA sequence itself

- There are at least two forms of information in the genome of the cell:
  - **Genetic information**: provides the building block for the manufacture of all proteins needed for the cell functional activity
  - **Epigenetic information**: provides additional instructions on how, when and where the genetic information should be used

# Chromatin organization

- Multiple levels of packing are required to fit the DNA into the cell nucleus
- Net result: each DNA molecule has been packaged into a mitotic chromosome that is 10,000-fold shorter than its extended length

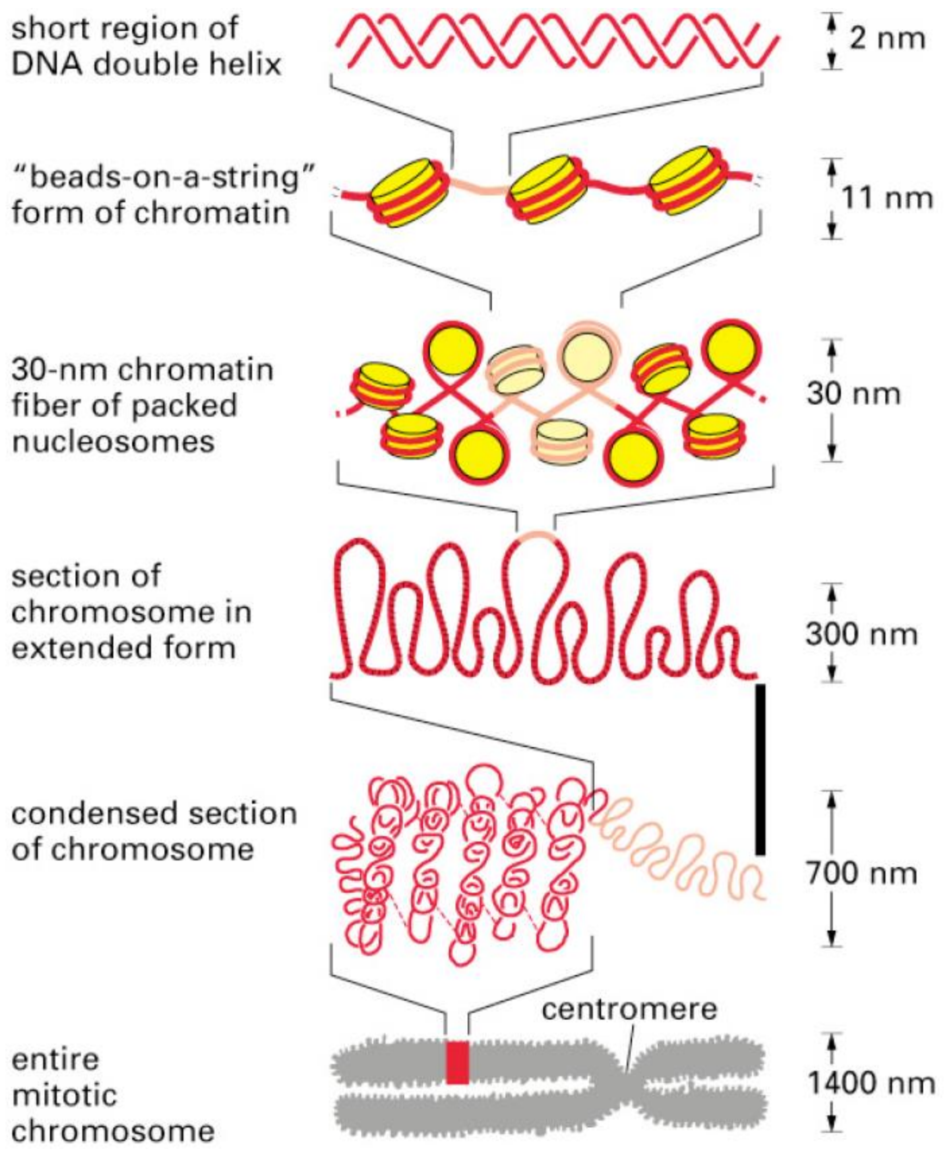



Fig. 4-55. Molecular Biology of the Cell (4th Ed.)

# Nucleosome

- The nucleosome consists of 146 bp of DNA wrapped around a protein core of 8 histones

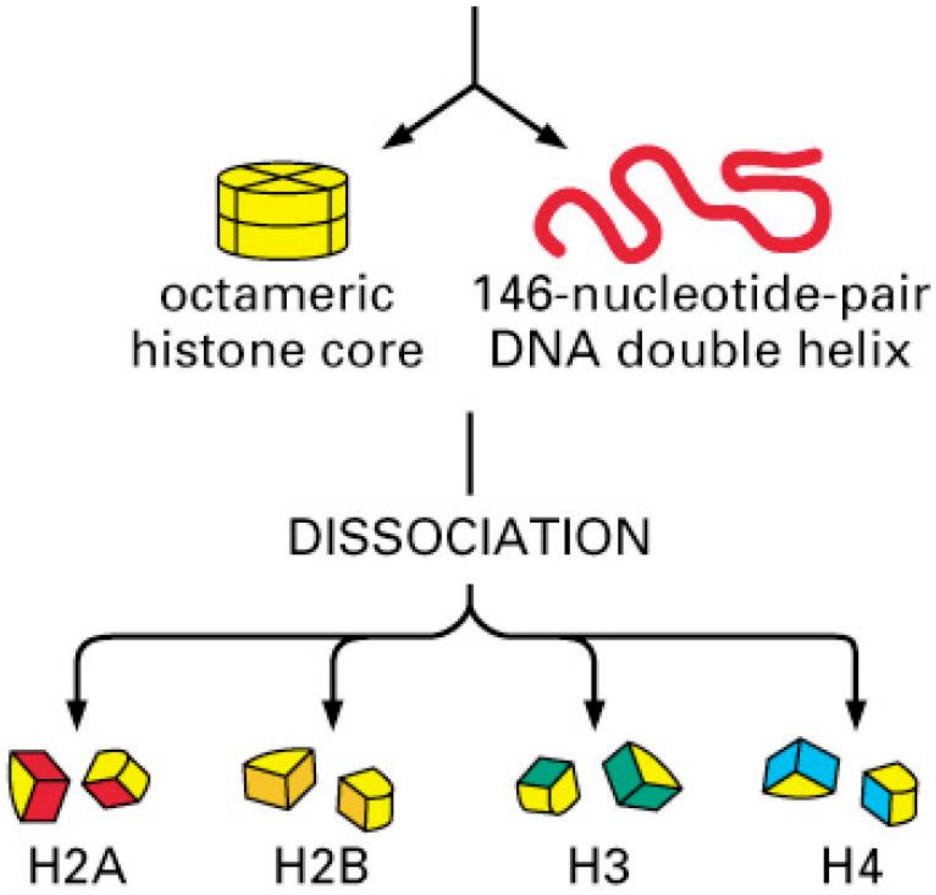



Fig. 4-24. Molecular Biology of the Cell (4th Ed.)

# Other DNA binding proteins

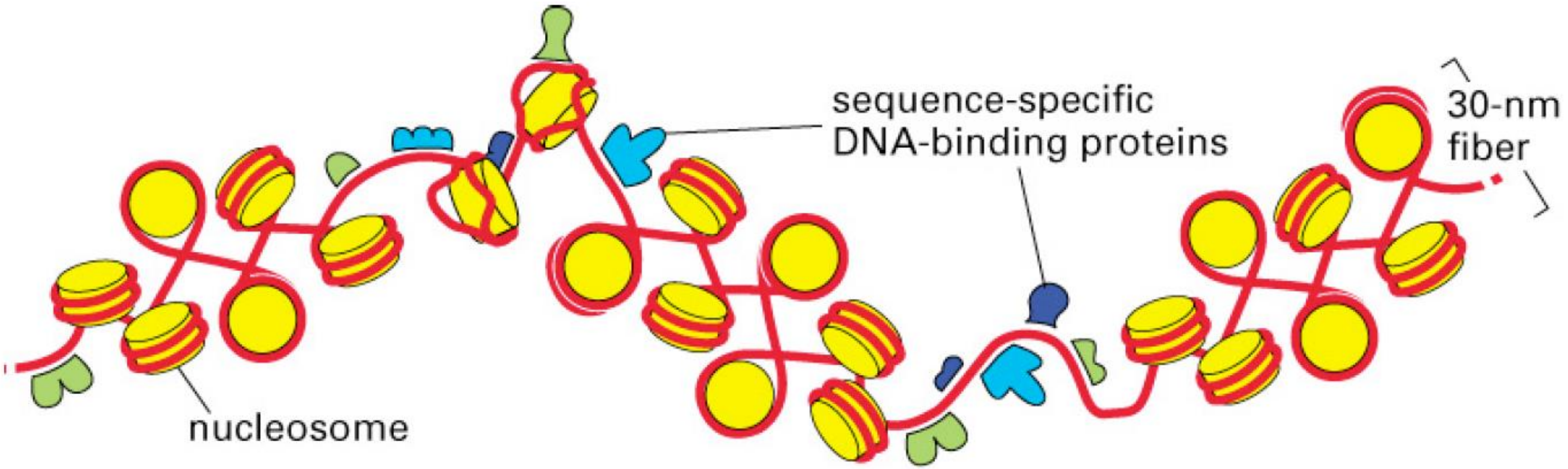



Fig. 4-30. Molecular Biology of the Cell (4th Ed.)

# Chromatin loops

- The 30 nm fiber is organized into loops that can be opened up individually

- This allows individual genes and sets of genes to be accessed without a global unpacking of the chromosome

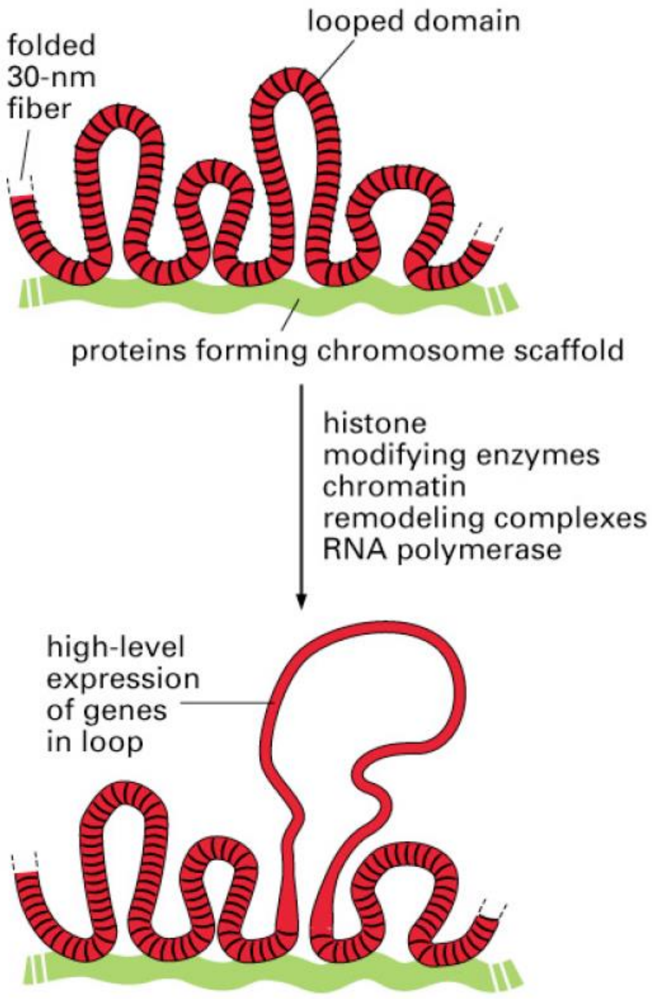


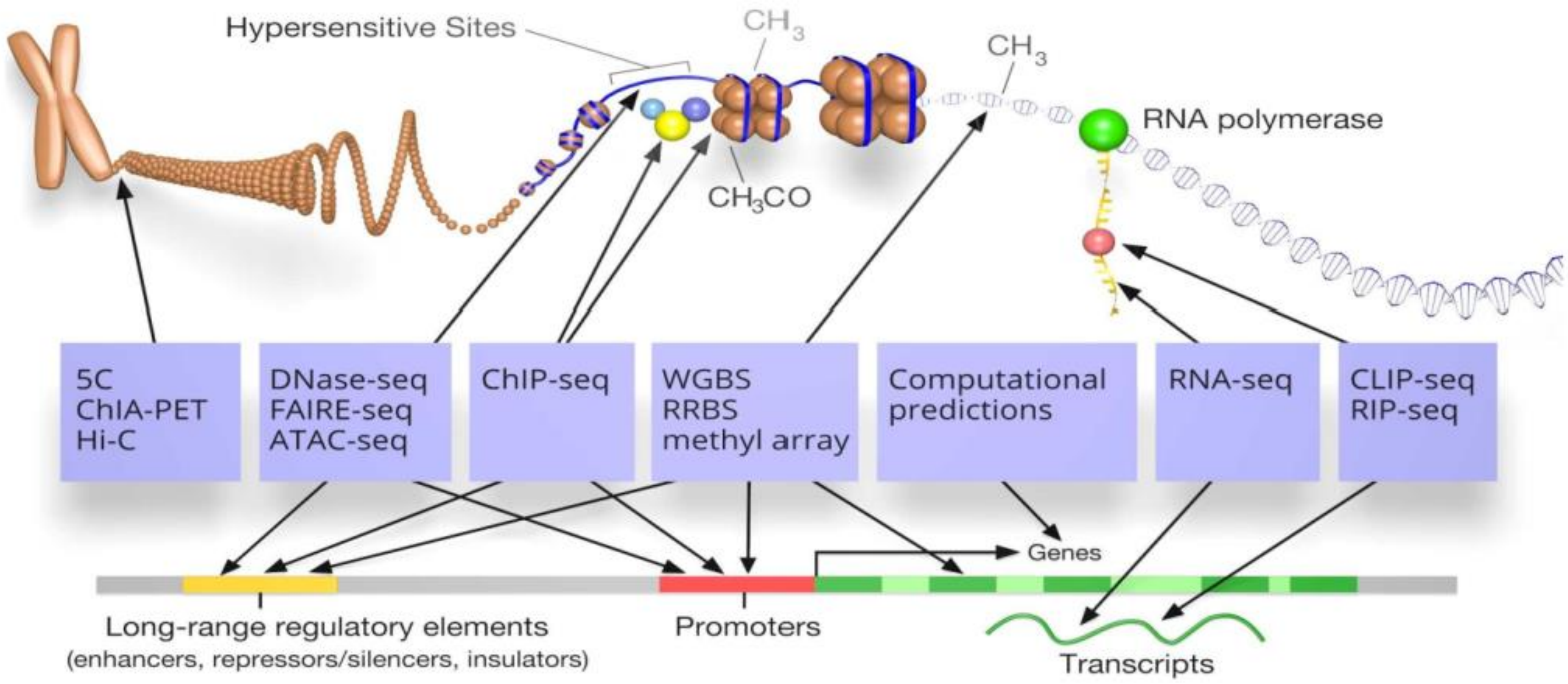

Fig. 4-44. Molecular Biology of the Cell (4th Ed.)

# Main epigenetic mechanisms



- **DNA methylation**: Methyl group (an epigenetic factor found in some dietary sources) can tag DNA and activate or repress genes
- **Histone modification**: The binding of epigenetic factors to histone "tails" alters the extent to which DNA is wrapped around histones and the availability of genes in the DNA to be activated
- **Non-coding RNA (ncRNA)**: A functional RNA molecule that is transcribed from DNA but not translated into proteins. In general, ncRNAs function to regulate gene expression at the transcriptional and post-transcriptional levels

# Sequencing for epigenetics

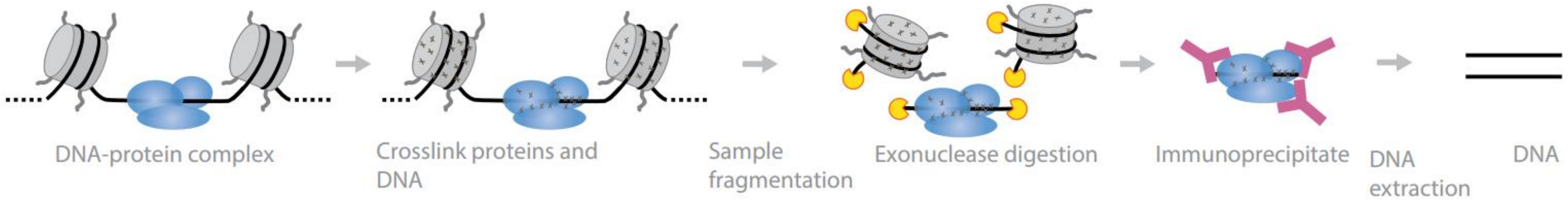



- Since DNA sequencing was first devised in 1977 by Frederick Sanger and his colleagues, sequencing methods have been evolved rapidly and volatile in sequencing objects (DNA, RNA, methylation, etc.)

# ChIP-seq

# ChIP-seq

- **Chromatin immunoprecipitation sequencing (ChIP-seq)** is used primarily to determine how transcription factors and other chromatin-associated proteins influence phenotype-affecting mechanisms
- **Goal:** Identify both direct and indirect protein-DNA interactions
- **Experimental approach:** Targeted chromatin pulldown after crosslinking followed by sequencing of captured DNA fragments



# Motivation for ChIP-seq

- General questions about gene (expression) regulation
  - Where are the relevant regulatory elements?
  - Which transcription factors (TFs) are involved?
  - What does chromatin state reveal?
  - What are the changes across conditions?

- To interpret and guide the studies of human genetics
  - Where are the relevant regulatory elements within the linkage block of GWAS signal or family studies?
  - What regulatory elements are likely to control critical developmental or phenotype-associated genes?
  - Guide resequencing and functional studies

# Strengths and limitations

- **Pros**
  - Base-pair resolution of protein-binding site
  - Can map specific regulatory factors or proteins
  - The use of exonuclease eliminates the contamination by unbound DNAs
- **Cons**
  - Nonspecific antibodies can dilute the pool of DNA-protein complexes of interest
  - The target protein must be known and able to raise an antibody

# General applications

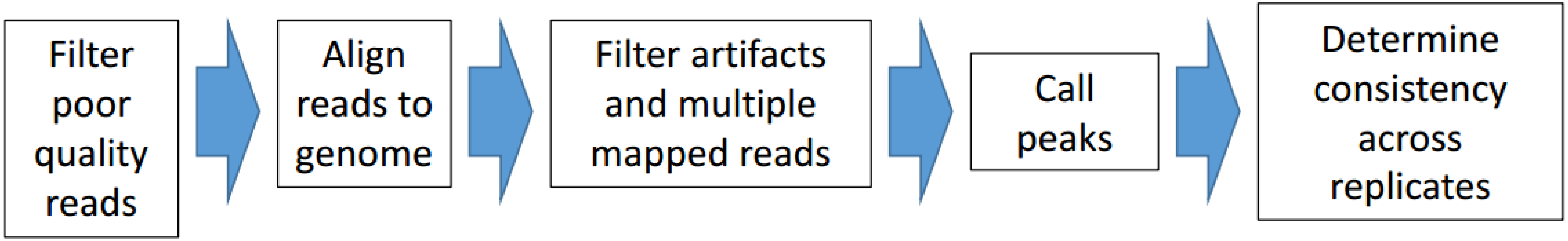


1. Map binding sites of a specific TF

2. Map interactions for other functional proteins
   - Co-activators (e.g. p300)
   - Structural proteins (e.g. CTCF)
   - Chromatin remodeling proteins (e.g. BRG1)
   - Nucleosome components (e.g. H2A)

3. Study chromatin regulation via histone modifications
   - **Annotation**: promoters, enhancers, gene bodies, etc.
   - **Function**: active enhancer, repressed chromatin, elongation, etc.

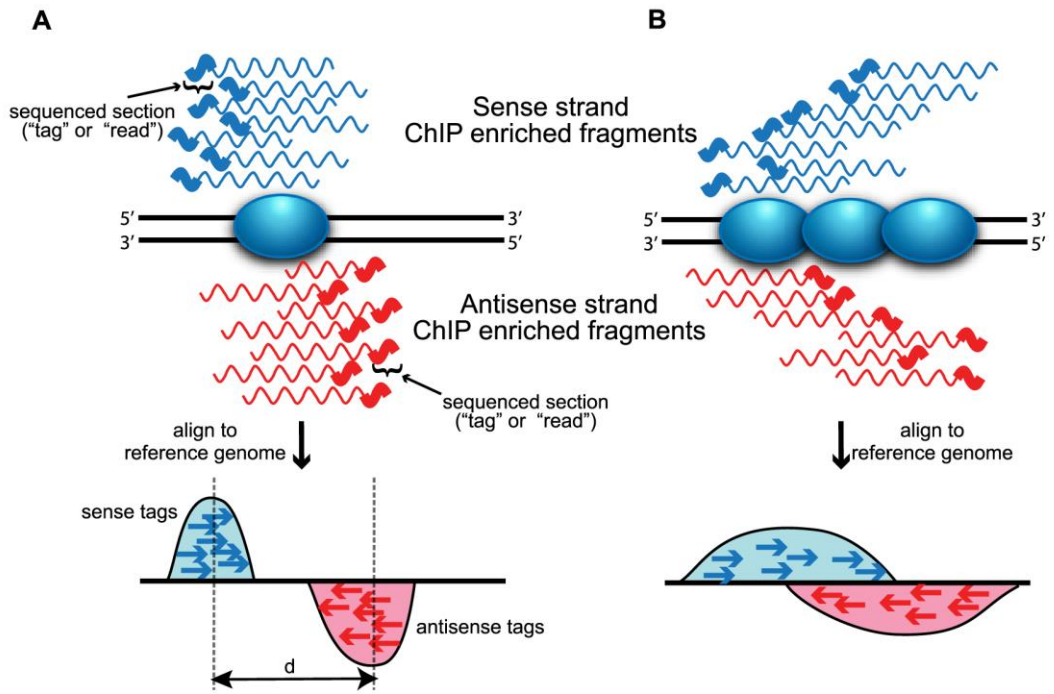
# ChIP-seq data analysis pipelines



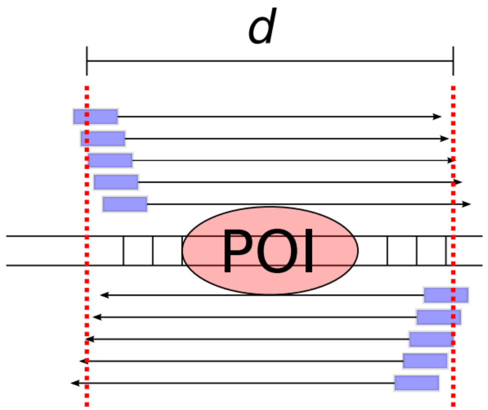

- **ENCODE ChIP-seq pipeline:** https://github.com/ENCODE-DCC/chip-seq-pipeline2
- **UC Davis ChIP-seq pipeline:** https://github.com/kernco/chipseq-pipeline

# Peak calling

- Peak calling algorithms help define the protein-DNA binding sites by identifying regions where sequence reads are enriched in the genome after mapping
- Why does the separation between peaks (d) correspond to the average sequenced fragment length?
  - The purple box shows the region of the fragment that is actually sequenced (often 36bp)
  - The entire fragment is longer, with the exact size depending on the experimental fragmentation protocol
  - On average, the protein of interest (POI) is located in the middle of the fragment
  - Therefore, the average distance between reads corresponds to the average fragment length

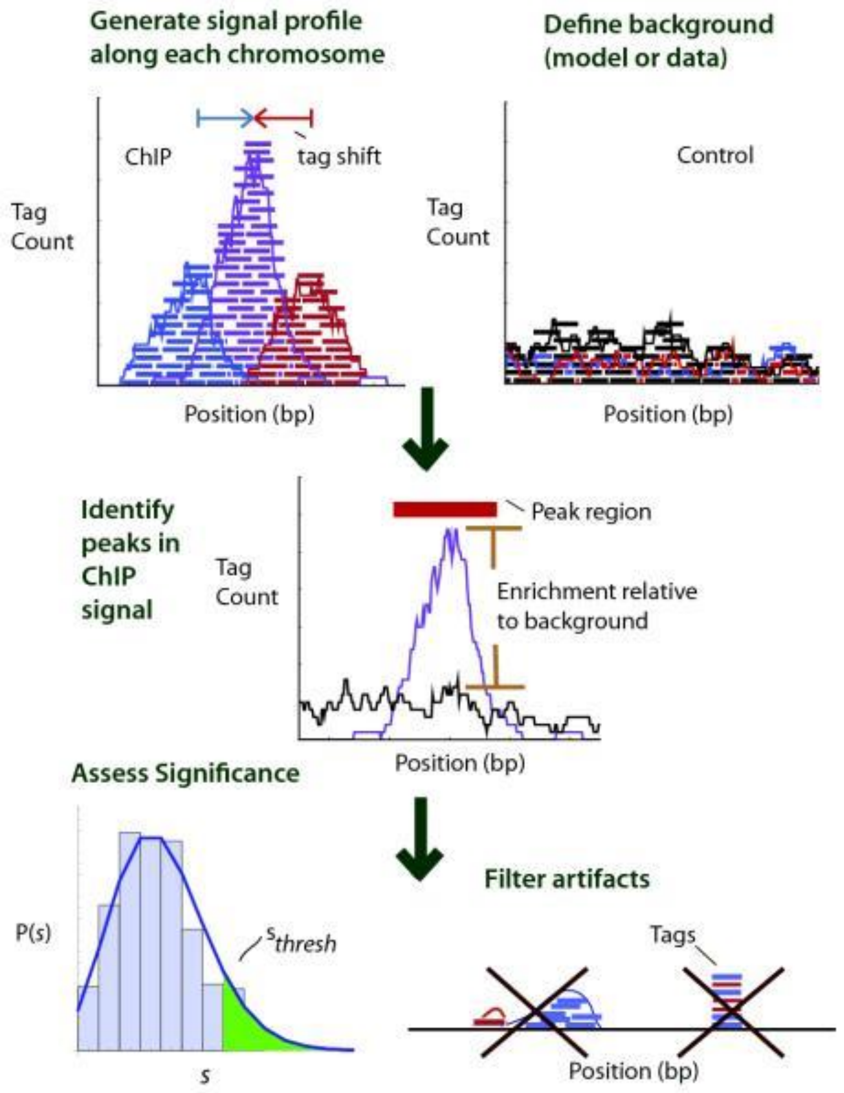



**The separation between peaks (d) corresponds to the average sequenced fragment length.** Wilbanks et al. PLOS ONE, 2010



# Peak evaluation

- Look for fold enrichment of the sample against input or expected background
- Estimate the significance of the fold enrichment using:
  - Poisson distribution
  - negative binomial (NB) distribution
  - background distribution from input DNA
  - model the background data to adjust for local variation (as in the MACS algorithm, described in Appendix A)

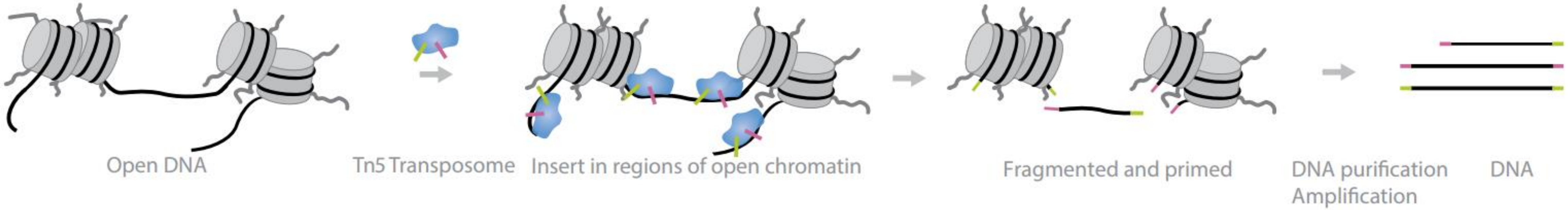



Pepke, S. et al., Nature Methods, 2009

# ATAC-seq

# ATAC-seq

- **Assay for Transposase-Accessible Chromatin using sequencing (ATAC-seq):** is based on direct *in vitro* transposition of sequencing adapters into native chromatin – as a rapid and sensitive method for integrative epigenomics analysis (Buenrostro, Jason D et al. Nature Method, 2013)
- **Experimental Approach:** DNA is incubated with Tn5 transposome, which performs adaptor ligation and fragmentation of open chromatin regions. Deep sequencing of the purified regions provides base-pair resolution of nucleosome-free regions in the genome





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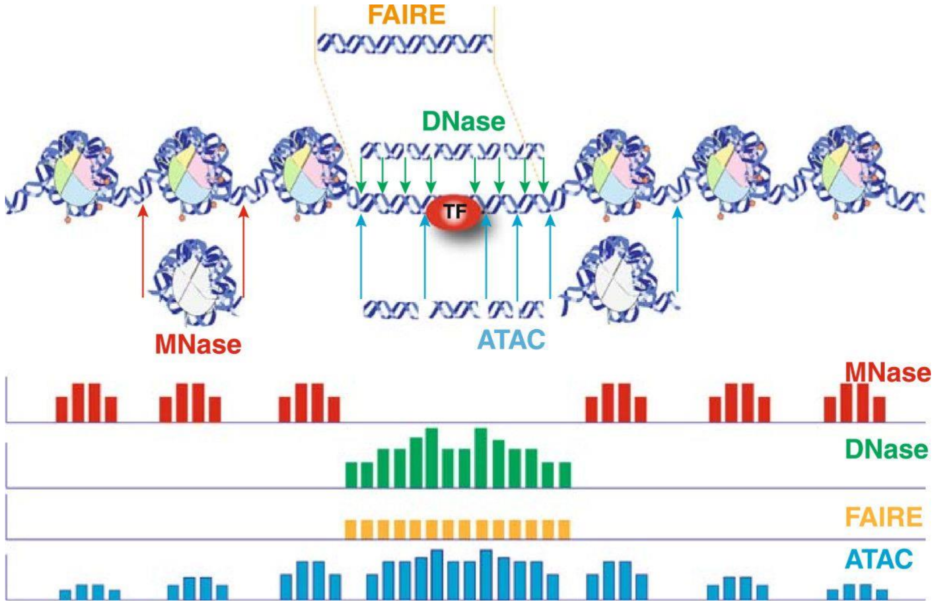
# Compare with related techniques

## Other techniques:

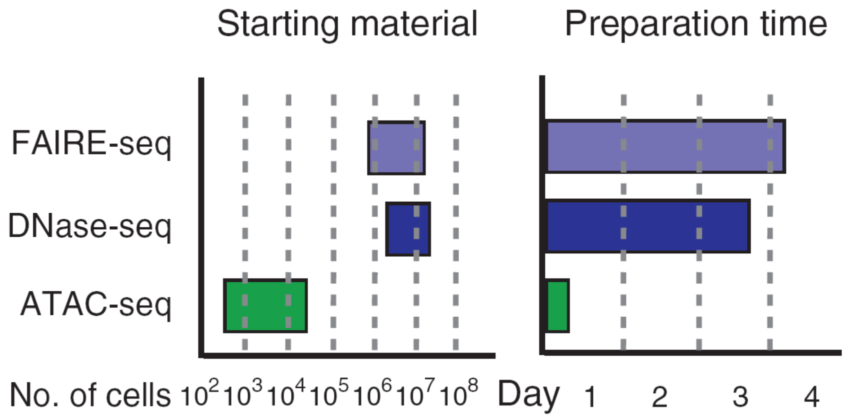
- MNase-seq: As MNase digests unprotected DNA, it is sensitive to open chromatin areas, and maps regions that are protected by nucleosomes (large open regions are digested)
- DNase-seq: As DNase I cuts within unprotected DNA, it maps open chromatin
- FAIRE-seq: Based on the phenol-chloroform separation of nucleosome-bound and free sonicated areas of a genome, it maps the open chromatin




Tsompana and Buck, Epigenetics & Chromatin, 2014

## Pros of ATAC-seq

- Sample demand is greatly reduced
  - The amount of cells required is from 500 to 50,000
  - Compared to the sample requirements of $10^7$cells in other experiments, ATAC-seq is friendly to the small amount of cells
- Reducing sequencing depth
- One-time access to chromatin open regions on the whole genome
- Very high signal-to-noise ratio compared to FAIRE-seq




JD Buenrostro et al., Nature Methods, 2013

# Workflow of ATAC-seq data processing

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Sequencing

↓

QC, read trimming — FastQC, Cutadapt, Trimmomatic etc.

↓

Alignment — BWA, Bowtie

↓

Peak calling — MACS2, Zinba etc.

↓

QC, data visualization — ChIPQC, IGV

↓

Downstream analysis

1. ATAC-seq can be further trimmed: estimating the percentage of sequence reads that map to the mitochondrial genome (the lower the better)
2. Use genome browsers to see raw tag density ---- UCSC, IGV, GIVE

**1. Full-read approach**

Bowtie2

Burrows-Wheeler Aligner (BWA)

**2. Chimeric alignment (unmapped from previous step, which may contains ligation junction)**

BWA

STAR

ChimeraScan



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# Hi-C

# Hi-C

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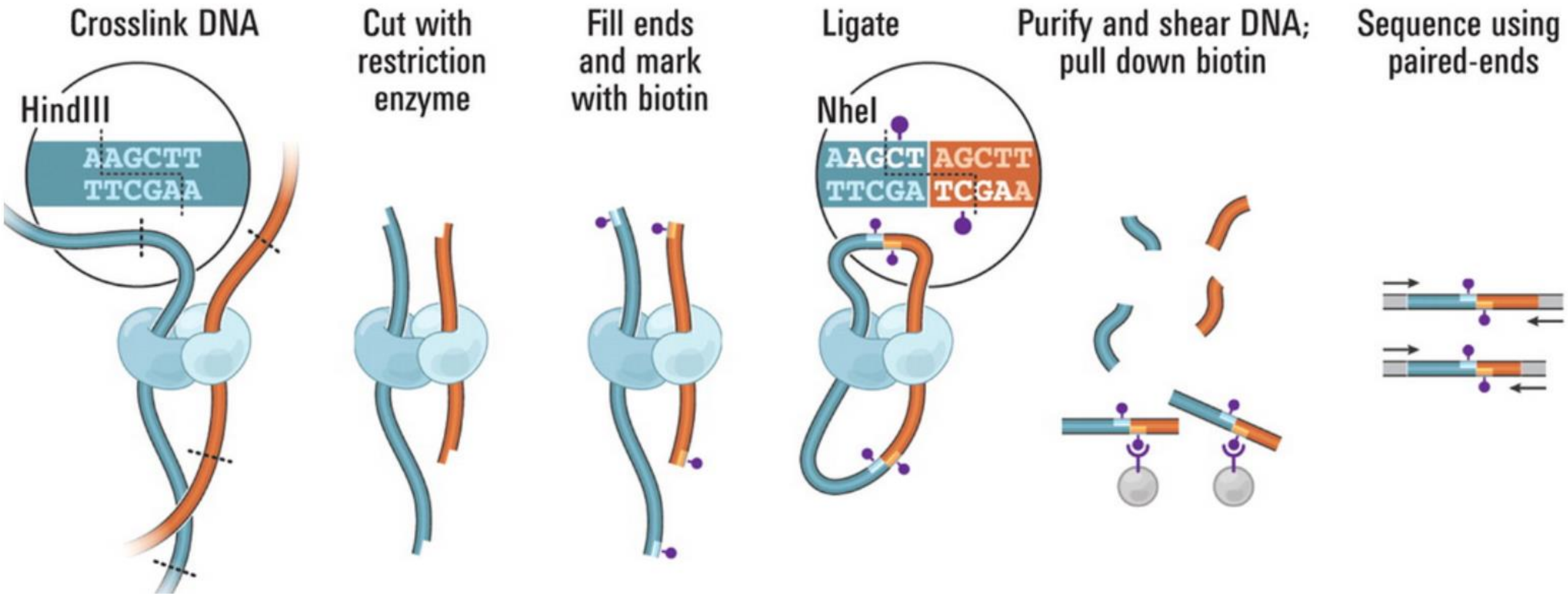
- Chromatin conformation capture sequencing (Hi-C) is used to analyze chromatin interactions (Lieberman-Aiden et al., Science, 2009; Rao et al., Cell, 2014)



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# Pros and cons of Hi-C

## Pros

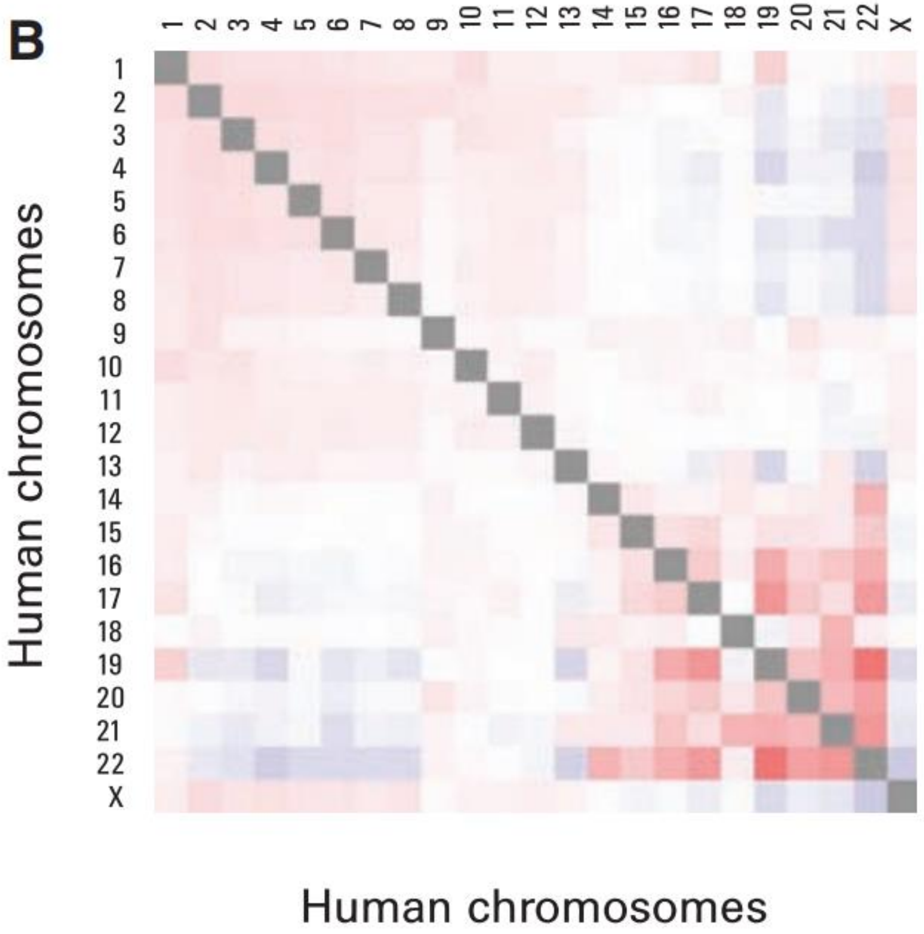
- Allows detection of long-range DNA interactions
- High-throughput method

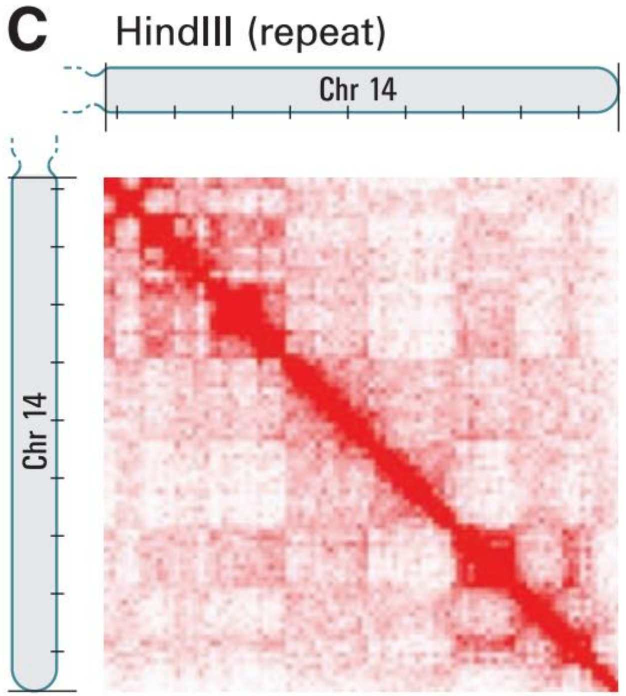
## Cons

- Detection may result from random chromosomal collisions
- Due to multiple steps, the method requires large amounts of starting material

# What can we learn from Hi-C methods?






- Cross-linking frequencies between sequences genome-wide
- Often referred to as "interactions"
- Visualized in contact map

Lieberman-Aiden et al., Science, 2009

# What can we learn from Hi-C methods?

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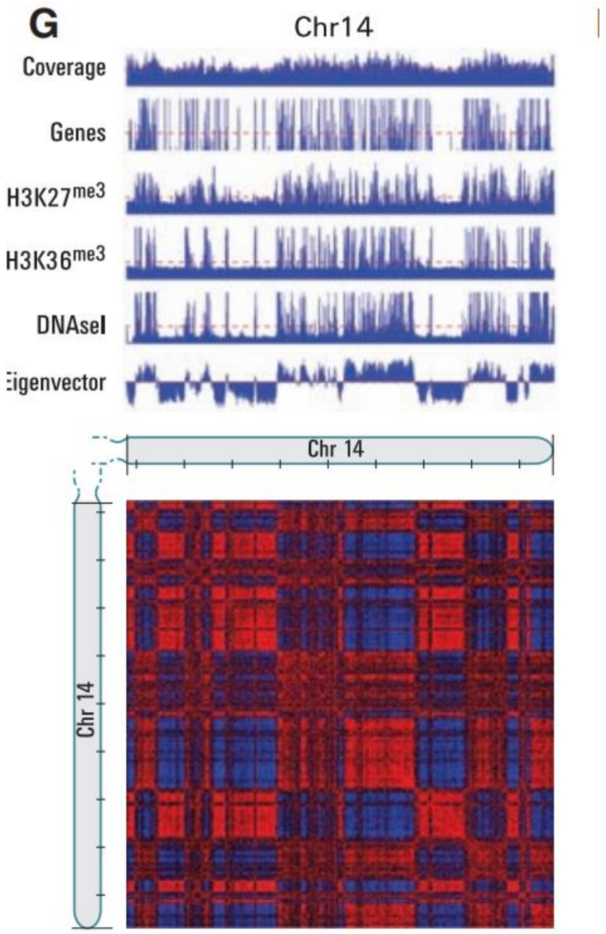
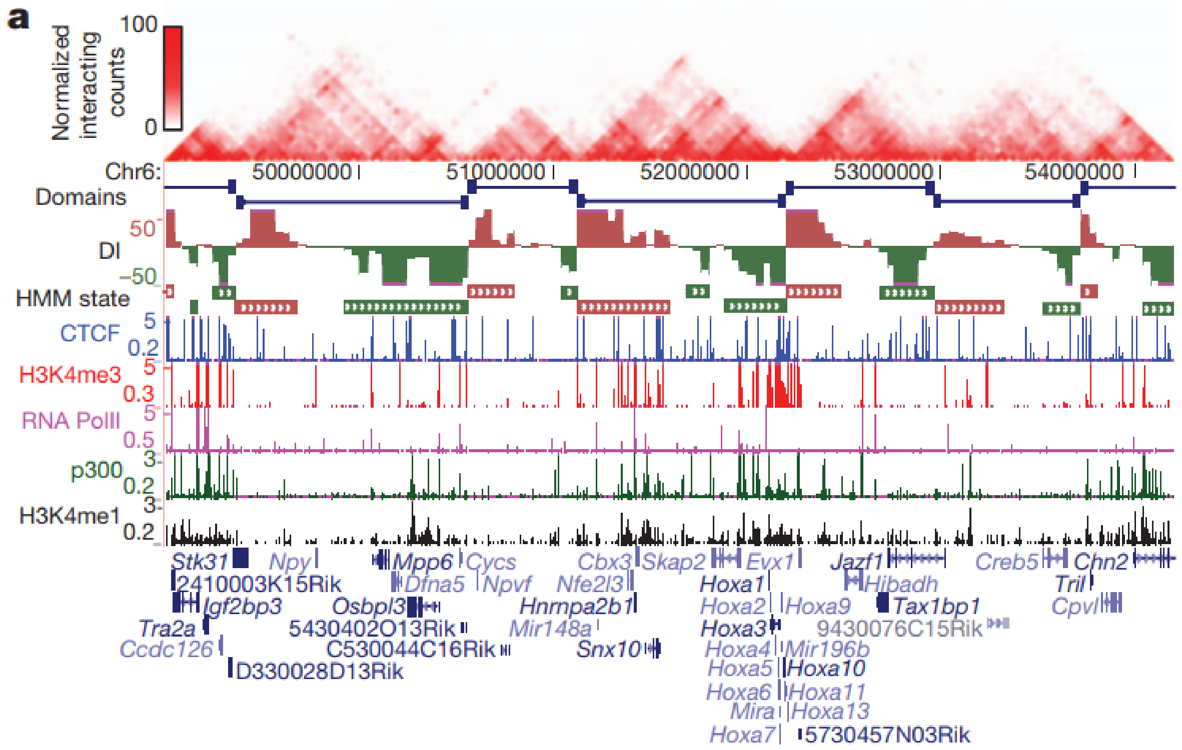
- **Chromosome compartmentalization**




Lieberman-Aiden et al., Science, 2009




Dixon et al., Nature, 2012



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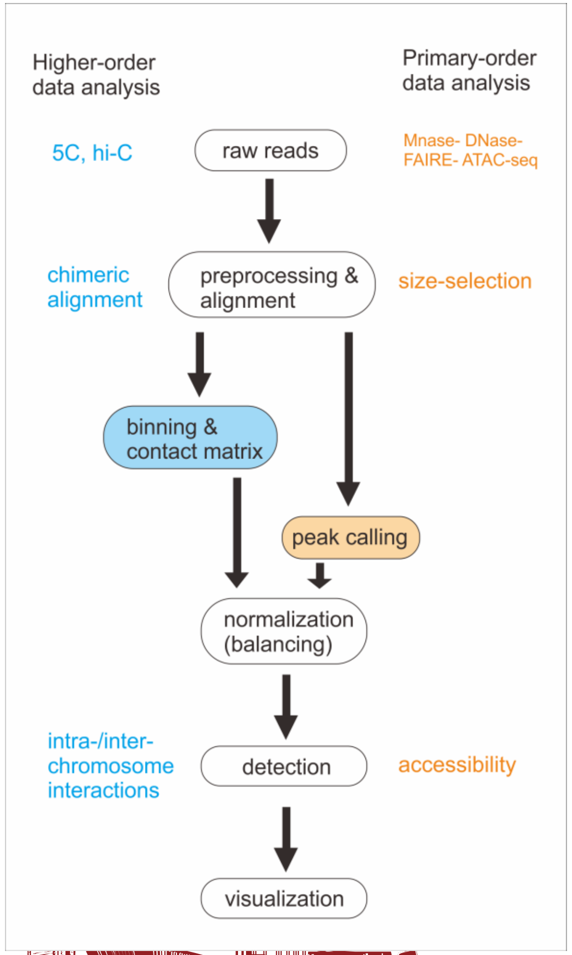
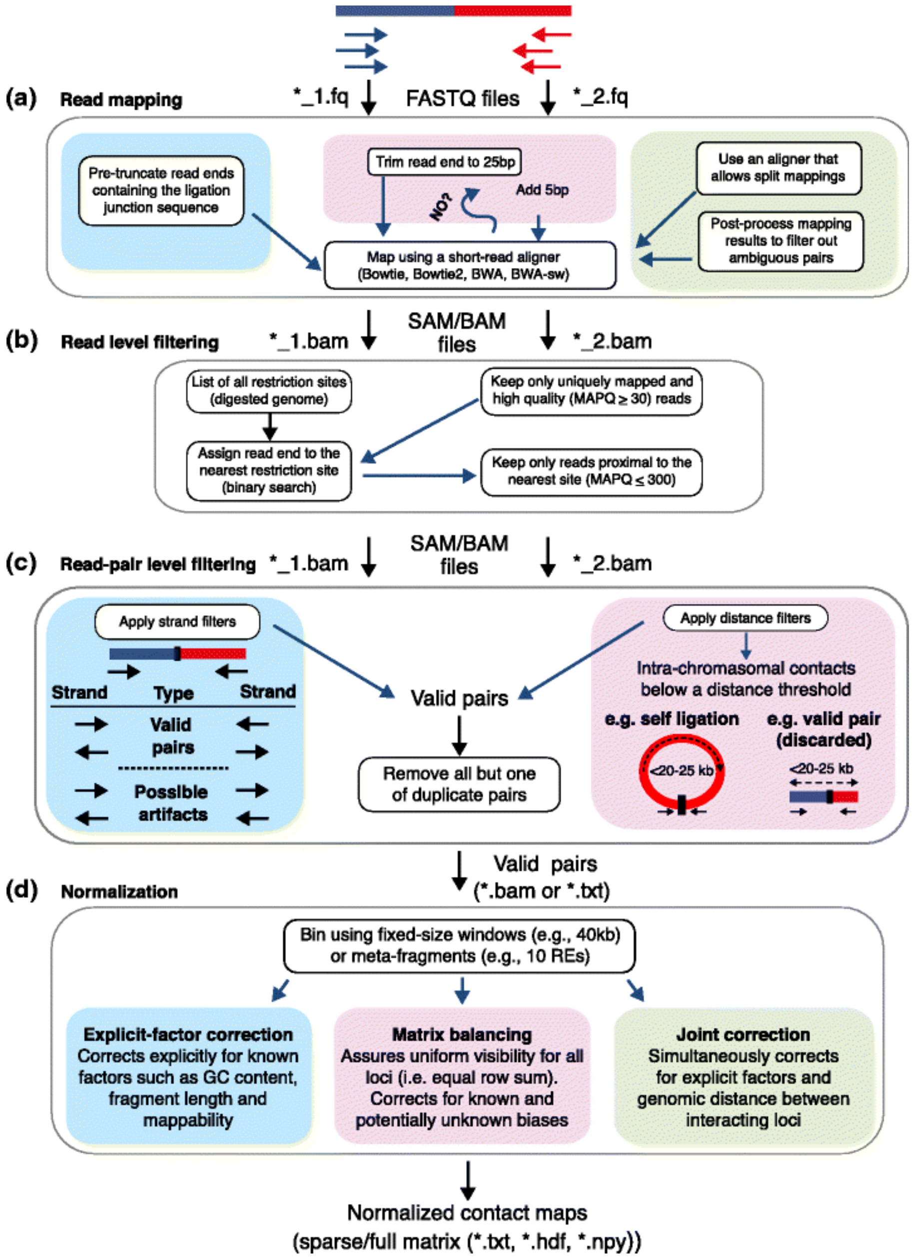
# Hi-C workflow

- Data analysis workflow for higher-order data and primary-order data
- The white box indicates common steps in the middle, and special analysis for different types are listed on two sides with colors




**Overview of Hi-C analysis pipelines**
Ay & Noble, Genome Biology, 2015

# Software tools for Hi-C data analysis

| Tool | Short-read aligner(s) | Mapping improvement | Read filtering | Read-pair filtering | Normalization | Visualization | Confidence estimation | Implementation language(s) |
|---|---|---|---|---|---|---|---|---|
| HiCUP [46] | Bowtie/Bowtie2 | Pre-truncation | ✓ | ✓ | – | – | – | Perl, R |
| Hiclib [47] | Bowtie2 | Iterative | ✓[a] | ✓ | Matrix balancing | ✓ | – | Python |
| HiC-inspector [131] | Bowtie | – | ✓ | ✓ | – | ✓ | – | Perl, R |
| HIPPIE [132] | STAR | ✓[b] | ✓ | ✓ | – | – | – | Python, Perl, R |
| HiC-Box [133] | Bowtie2 | – | ✓ | ✓ | Matrix balancing | ✓ | – | Python |
| HiCdat [122] | Subread | –[c] | ✓ | ✓ | Three options[d] | ✓ | – | C++, R |
| HiC-Pro [134] | Bowtie2 | Trimming | ✓ | ✓ | Matrix balancing | – | – | Python, R |
| TADbit [120] | GEM | Iterative | ✓ | ✓ | Matrix balancing | ✓ | – | Python |
| HOMER [62] | – | – | ✓ | ✓ | Two options[e] | ✓ | ✓ | Perl, R, Java |
| Hicpipe [54] | – | – | – | – | Explicit-factor | – | – | Perl, R, C++ |
| HiBrowse [69] | – | – | – | – | – | ✓ | ✓ | Web-based |
| Hi-Corrector [57] | – | – | – | – | Matrix balancing | – | – | ANSI C |
| GOTHiC [135] | – | – | ✓ | ✓ | – | – | ✓ | R |
| HiTC [121] | – | – | – | – | Two options[f] | ✓ | ✓ | R |
| chromoR [59] | – | – | – | – | Variance stabilization | – | – | R |
| HiFive [136] | – | – | ✓ | ✓ | Three options[g] | ✓ | – | Python |
| Fit-Hi-C [20] | – | – | – | – | – | ✓ | ✓ | Python |

Ay & Noble, Genome Biology, 2015

# Biological Background of Proteomics

# What is proteomics?

- The **proteome** is the entire set of proteins that is produced or modified by an organism or system
    - The term of "proteomics" was coined in 1994 by Marc Wilkins, blending "protein" and "genome"
- **Proteomics** is the large-scale study of proteins
    - The term was coined in 1997, in analogy to "genomics"
    - Proteomics has enabled the identification of ever increasing numbers of proteins
    - It is an interdisciplinary domain that covers all levels of proteins, e.g. protein composition, structure and activity
    - Important for functional genomics and systems biology

# Approaches to protein identification

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- There are 3 main approaches to protein identification
  - Direct protein sequencing
  - Gel eletrophoresis
  - Mass spectrometry

- High-throughput protein sequencing is to deduce the amino acid sequence of a protein. It is still very difficult
- Currently, research is focused on peptide sequencing, i.e. getting the amino acid sequence of a short fragment of a protein (of length ≈ 10)



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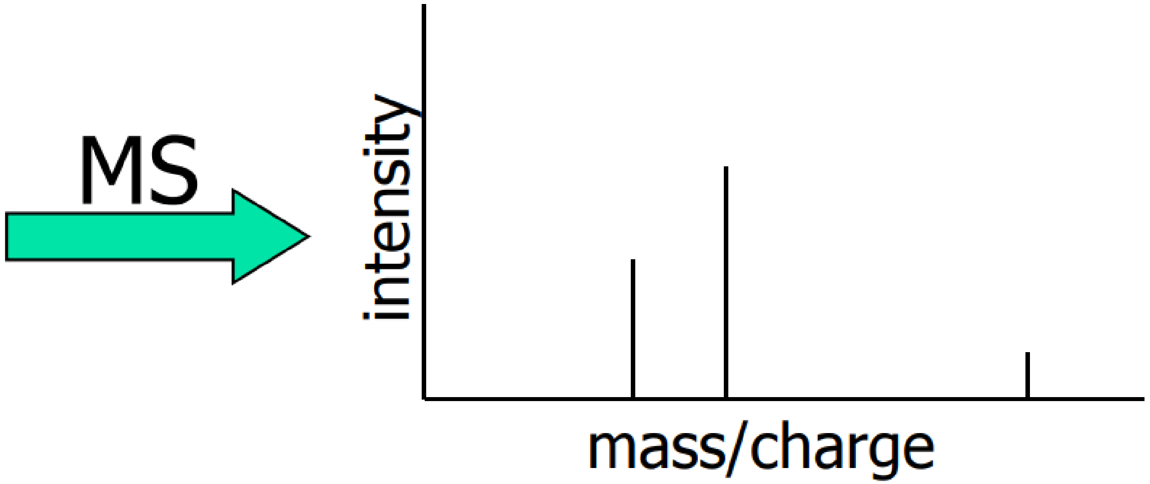
# Mass spectrometry



- Key for deducing the peptide sequence: mass
- Mass Spectrometry is a machine which can separate and measure samples with different mass/charge ratios
- Example:

Sample 1: m/z=100Da, 10mol
Sample 2: m/z=50Da, 50mol
Sample 3: m/z=33Da, 30mol




Dalton(Da) is a mass unit. E.g. H is of mass 1Da

# History



- Peptide sequencing was invented by Pehr Edman (1949) and Frederick Sanger (1955)
- In 1966, Biemann et al. successfully sequenced a peptide using a mass spectrometer machine
- During 1980s, sequencing using mass spectrometry becomes popular

# Amino acid residue mass

- Observations from the table:
  - Amino acid residue = amino acid losing a water
  - I and L have the same mass
  - Smallest mass is G (57.05 Da)
  - Largest mass is W (186.21 Da)
  - Most of the amino acids have different masses. Thus, with high chance, different peptides have different masses
- The mass given by a mass spectrometer has a max. error $\pm 0.5\mathrm{Da}$. It can separate most of the peptides

| | | | |
|---|---|---|---|
| A | 71.08 | M | 131.19 |
| C | 103.14 | N | 114.1 |
| D | 115.09 | P | 97.12 |
| E | 129.12 | Q | 128.13 |
| F | 147.18 | R | 156.19 |
| G | 57.05 | S | 87.08 |
| H | 137.14 | T | 101.1 |
| I | 113.16 | V | 99.13 |
| K | 128.17 | W | 186.21 |
| L | 113.16 | Y | 163.18 |

# Protein identification process

- **Input**: A protein sample

A) Biology part:

- 1. Digest the protein into a set of peptides
- 2. By HPLC + Mass Spectrometer, separate the peptides
- 3. Select a particular peptide
- 4. Fragment the selected peptide
- 5. Get the tandem mass (MS/MS) spectrum of the selected peptide

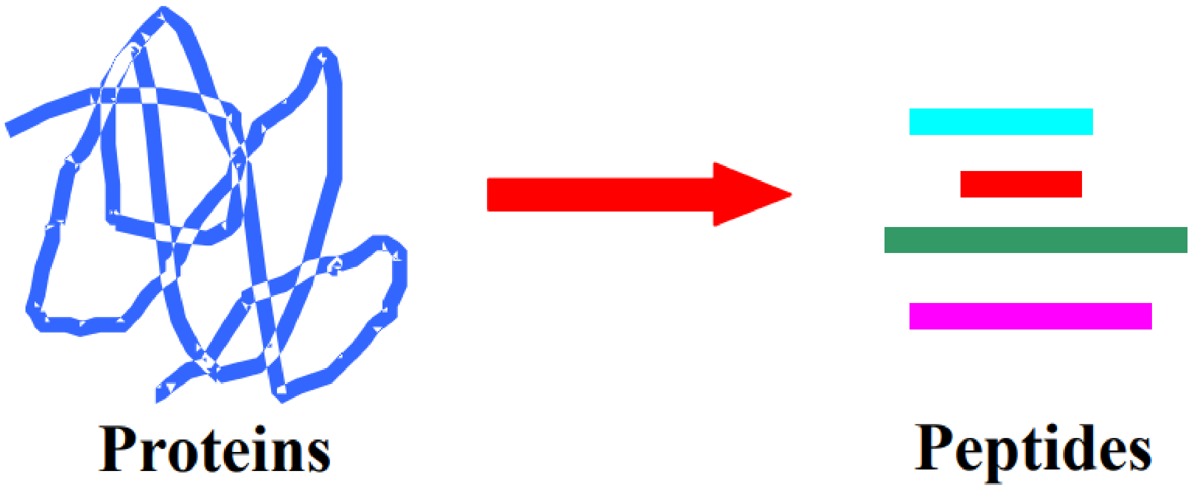
B) Computing part:

- De novo peptide sequencing
- Protein database search

# Digest a protein into peptides



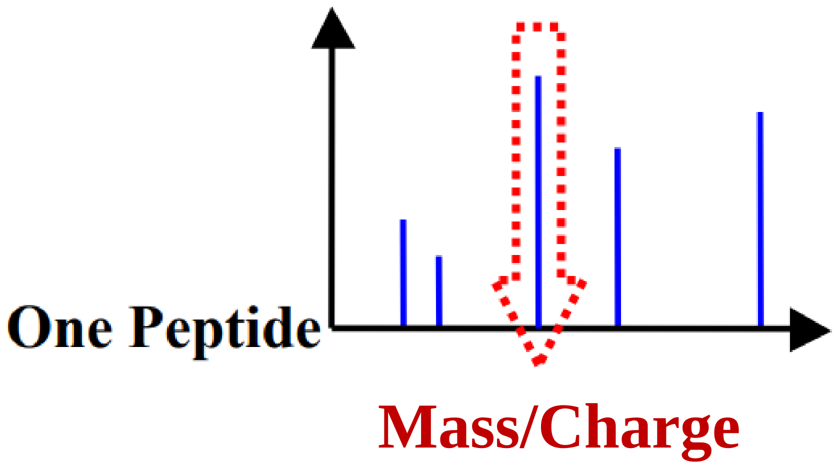
- By an enzyme, digest a protein into short peptides
- If we digest a protein using trypsin,
  - It digests the protein at K or R that are not followed by P
  - After digestion, we will get a set of peptides ending with K or R
- E.g. ACCHCKCCVRPPCRCA -> ACCHCK, CCVRPPCR



# Selecting a particular peptide



- **HPLC** (High Performance Liquid Chromatography) can separate a set of peptides in a high pressure liquid chromatography
- After HPLC, the mixture of peptides are analyzed by MS. Then, we get the MS spectrum



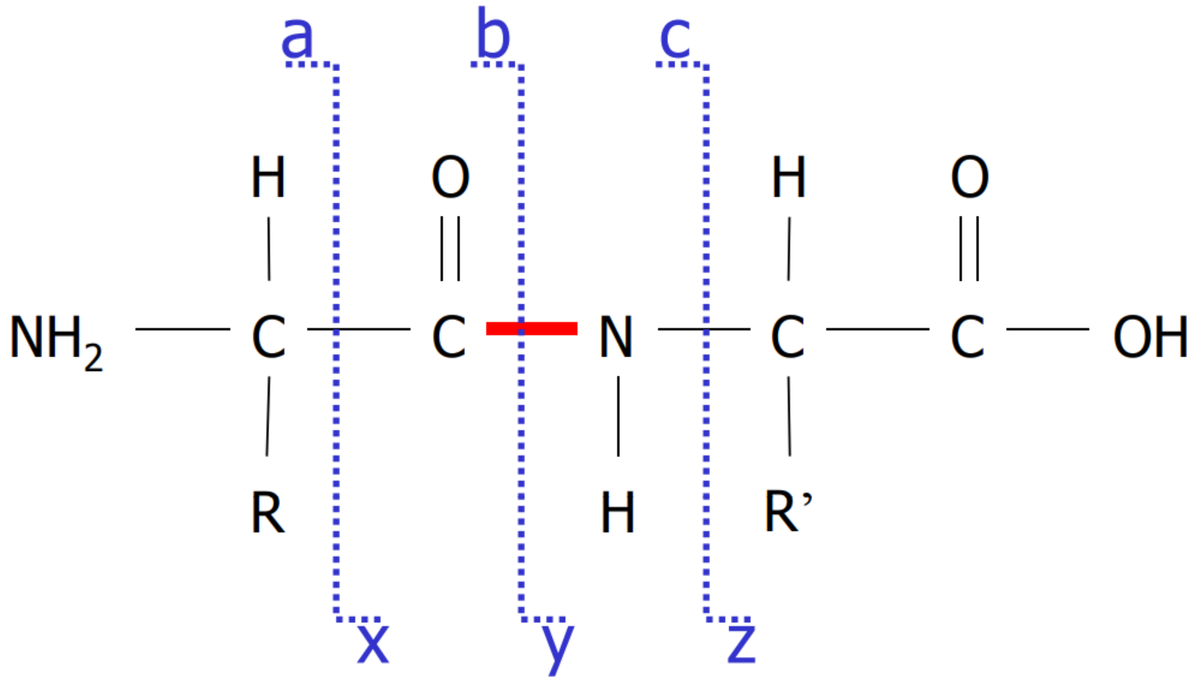

- The peptide of a particular mass is selected

# Fragmentation of peptide (I)

- Fragmentation tries to break the selected peptide at all positions in the peptide background
- Usually, fragmentation is by Collision Induced Dissociation (CID)
  - The peptide is passed into the collision cell, which has been pressurized with argon (i.e. inert gas)
  - Collision between peptide and argon break the peptide
- Each peptide is usually fragmented into 2 pieces
  - Prefix fragment and suffix fragment (either one fragment will be charged but not both)

# Fragmentation of peptide (II)

- Most often, the peptide is broken at C-C, C-N, N-C bonds
  - Result in a-ions, b-ions, c-ions, x-ions, y-ions, and z-ions
- Based on experiment
  - The intensity of y-ions is larger than that of b-ions
  - The intensities of other ions are even smaller



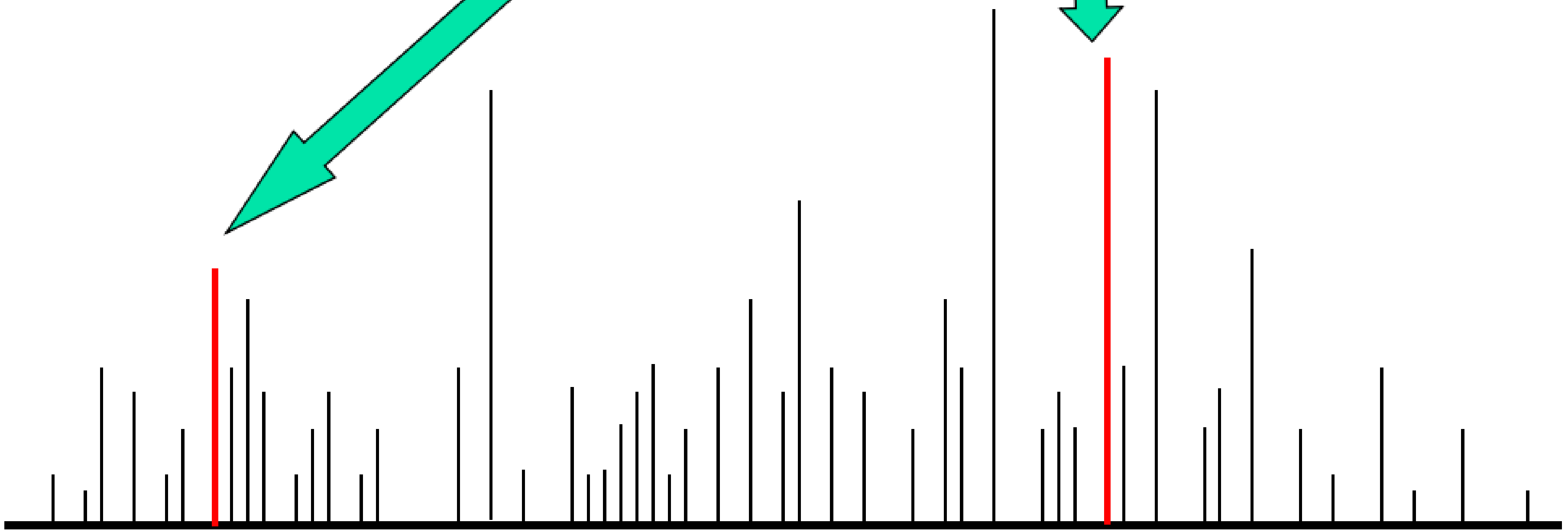
# Fragmentation of peptide (III)

r = w(CTVFT)
w = w(CTVFTEPREFK)

CTVFTEPREFK

fragmentation

CTVFT EPREFK



r+1 (mass of b-ion)

w-r+19 (mass of y-ion)

# Mass of the ions (I)



- Let A be the set of amino acids. For every $a \in A$, $w(a)$ = mass of its residue
- Let $P = a_1a_2...a_k$ be a peptide
  - $w(P) = \sum_{1 \le j \le k} w(a_j)$
- Actual mass of the peptide with sequence P is
  - $w(P)+18$ (since it has an extra $H_2O$)
- Mass of b-ion of the first i amino acids is
  - $b_i = 1 + w(a_1a_2...a_i)$
- Mass of y-ion of the last (k – i) amino acids is
  - $y_{i+1} = 19 + w(a_{i+1}...a_k)$
- Note: $b_i + y_{i+1} = 20 + w(P)$

# Mass of the ions (II)



- E.g. a peptide P = **SAG**
  - w(P) = w(S) + w(A) + w(G) = 215.21
  - Actual mass of P = w(P) + 18 = 233.21
  - $y_1$ = w(SAG) + 19 = 234.21
  - $y_2$ = w(AG) + 19 = 147.13
  - $y_3$ = w(G) + 19 = 76.05
  - $b_1$ = w(S) + 1 = 88.08
  - $b_2$ = w(SA) + 1 = 159.16
  - $b_3$ = w(SAG) + 1 = 216.21

# Other ion types

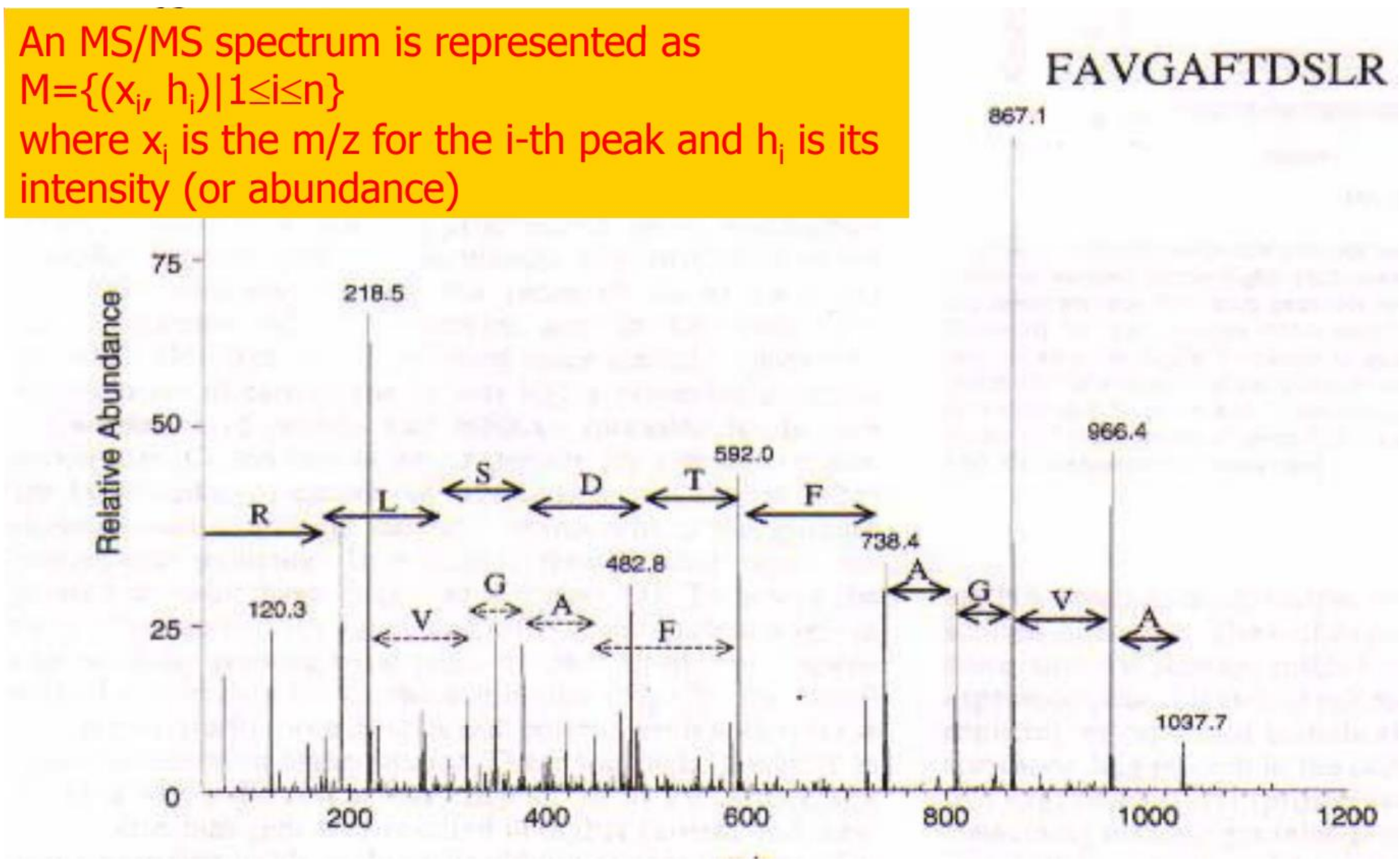
- Apart from a-ion, b-ion, c-ion, x-ion, y-ion, and z-ion, there are also variations with additional loss of
  - A water molecule
  - An ammonia molecule
  - A water and an ammonia molecule
  - Two water molecules

- E.g. y-$H_2O$, y-$NH_3$, y-$H_2O$-$H_2O$, y-$H_2O$-$NH_3$

# Tandem mass spectrum (MS/MS Spectrum)



An MS/MS spectrum is represented as
$M=\{(x_i, h_i)|1 \le i \le n\}$
where $x_i$ is the m/z for the i-th peak and $h_i$ is its intensity (or abundance)



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# Computational problems

- There are mainly 3 computational problems
  - 1. De novo peptide sequencing
  - 2. Peptide identification through database search
  - 3. Identification of PTM (post-translational modification)

- We will discuss problems 1 and 2

# Online resources for proteomics

- ExPASy at SIB (Swiss Institute of Bioinformatics):
  - https://expasy.org
  - A Bioinformatics Resource Portal providing access to scientific databases and software tools in different areas of life science
  - Proteomics is a foremost category (https://www.expasy.org/proteomics)
- PRIDE at EBI: A database for mass spectrometry
  - http://www.ebi.ac.uk/pride/
  - A centralized, standards compliant, public data repository, including information on protein and peptide identifications, PTMs, and supporting spectral evidence
- Other repositories:
  - Global Proteome Machine Database (GPMDB): (http://gpmdb.thegpm.org/index.html)
  - PeptideAtlas (http://www.peptideatlas.org)
  - MOPED (Model Organism Protein Expression Database): (http://moped.proteinspire.org)
  - Tranche (http://www.proteomexchange.org/databases/tranche)

# De Novo Peptide Sequencing

# De novo peptide sequencing problem



- **Input**:
  - An MS/MS spectrum M
  - The total mass wt of the peptide
  - An error bound δ (default δ = 0.5)

- **Output**:
  - The peptide sequence

# Assumption of the spectrum

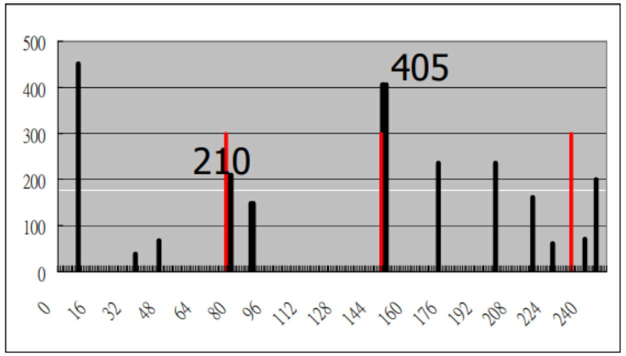
- We assume that all the ions are singly charged
- In fact, in an MS/MS experiment, an ion can be charged differently
- Fortunately, if a spectrum has peaks corresponding to multiply charged ions, there exists standard method to convert those peaks to their singly charged equivalents

# Simple scoring scheme

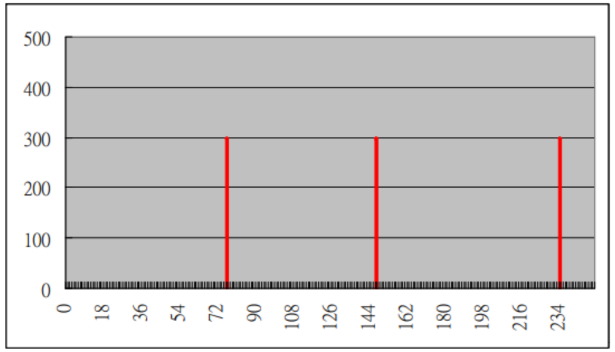
- Consider a peptide P = $a_1 a_2 \ldots a_k$
  - Recall that y-ions are expected to have the highest intensities
  - If M is a spectrum for P, we can find peaks for m/z = $y_i$ for $i$ = 1, 2, …, $k$
- So, we define the score function as:

$$\text{score(M, P)} = \sum\{h \mid (x, h) \in M, |x - y_i| \le \delta \text{ for } i = 1,2,\ldots,k\}$$

- Example: P = SAG
  - $y_1$ = 57.05 + 71.08 + 87.08 + 19 = 234.21
  - $y_2$ = 57.05 + 71.08 + 19 = 147.13
  - $y_3$ = 57.05 + 19 = 76.05
- score(M, P) = 210 + 405 = 615




Black peaks: real peaks




Red peaks: artificial y-ions

# Refined problem

- **Input**:
  - An MS/MS spectrum $M$
  - The total mass $wt$ of the peptide
  - An error bound $\delta$

- **Output**:
  - A peptide $P$ such that $wt - \delta \le w(P) \le wt + \delta$, which maximizes score($M$, $P$)

# Brute-force solution

- For every possible peptide P such that $|w(P) - wt| \le \delta$, compute score(M, P)
- Report the peptide P such that:
  - $| w(P) - wt | \le \delta$, and
  - score(M, P) is maximized

- The above algorithm takes **exponential time**. Very slow!
- Can we solve the problem faster? Yes, by dynamic programming (PEAKS algorithm in Appendix B)

# Protein Database Search

# Protein database searching problem

- **Input:**
  - A database of protein (DB)
  - A raw MS/MS spectrum (M)
  - The mass wt of the peptide corresponding to M
- **Output:**
  - A protein whose peptide is expected to have mass wt and a MS/MS spectrum similar to M

- In this lecture, let’ s look at a solution called SEQUEST (Eng et al. 1994)

# SEQUEST



- **Step 1**: Reduction of the tandem mass spectrometry data
  - To avoid noise, only 200 most abundant signals of the raw spectrum are used
  - Also, the total signals of the 200 signals are renormalized to 100
- **Step 2**: Search the protein database DB to find all peptides such that each peptide P has mass within (wt $\pm 1$) Da

# SEQUEST

- **Step 3**: Rank the top 500 fit sequences by a specific scoring function
- **Step 4**: Compare the spectral similarity. Use cross-correlation analysis to generate the final score and rank the sequences

- The abundance of ions in the hypothetic spectrum:
  - 50 (b-ion, y-ion), 25 (mass/charge within $\pm$ 1 from b or y), or 10 (a-ion)

# Other tasks of proteomics

- Protein Quantification
- Analysis of protein post-translational modifications (PTMs)
- Protein localization analysis
- Protein functional analysis
- Etc.

# Recap



- In this lecture, we have learned:
  - Basics of epigenomic mechanisms
    - ChIP-seq
    - ATAC-seq
    - Hi-C
  - Basics of proteomics techniques and related computational methods:
    - Basic mechanisms of mass spectrometry for protein identification
    - Algorithm of SEQUEST for protein database search with raw mass spectrometry data

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# Appendix A: MACS Algorithm

# MACS

- MACS is one of the most popular peak calling software. It was developed by Shirley Xiaole Liu's lab at Harvard University

## Core idea of the algorithm

- The binding of TFs (transcription factors) to genome is a relatively random process, i.e. every position on the genome have the chance to be seen by a TF (however, with a different probability). Peak calling is aimed to find those hot spots that are easily seen
- It uses a Poisson distribution to assign p-values to peaks. But the distribution has a dynamic parameter λ, i.e. the event rate, to capture the influence of local biases

$$P(k \text{ events in interval}) = e^{-\lambda}\frac{\lambda^k}{k!}$$

- Here $\lambda = np, p = \frac{l}{s}$, $n$ is the number of reads from sequencing, $l$ is length of a single read, and $s$ is length of the whole genome

# Sliding windows

- Given a sonication size (bandwidth) and a threshold of high-confidence fold-enrichment (mfold), MACS slides a window of length 2×bandwidth across the genome to find regions with tags more than mfold enriched relative to a random tag genome distribution

- Two parameters:
  - mfold specifies an interval of high-confidence enrichment ratio against the background on which to build the model. The default value in (10, 30) means that a model will be built on the basis of regions having read counts that are 10- to 30-fold of the background
  - bandwidth is half of the sliding window size used in the model-building step. It is set according to the length of the fragments expected experimentally from the sonication procedure

# Estimate fragment size

**Algorithm 1** Estimate Fragment Size

1: `Slide a window of` $2 \times \text{bandwidth}^3$ `across genome`
2: `Identify regions of moderate enrichment (mfold: 10-30 fold)`
3: **for each** peak $i$ of 1000 randomly chosen enriched regions **do**
4: `separate reads into + and - strand`
5: `Calculate mode of + and - summit`
6: $d_i \leftarrow |\text{mode}_+ - \text{mode}_-|$
7: **end for**
8: $d \leftarrow \text{average}_i(d_i)$

Thus, the distance between bimodal summits is assumed to be the estimated DNA fragment size *d*

# Strengths of MACS

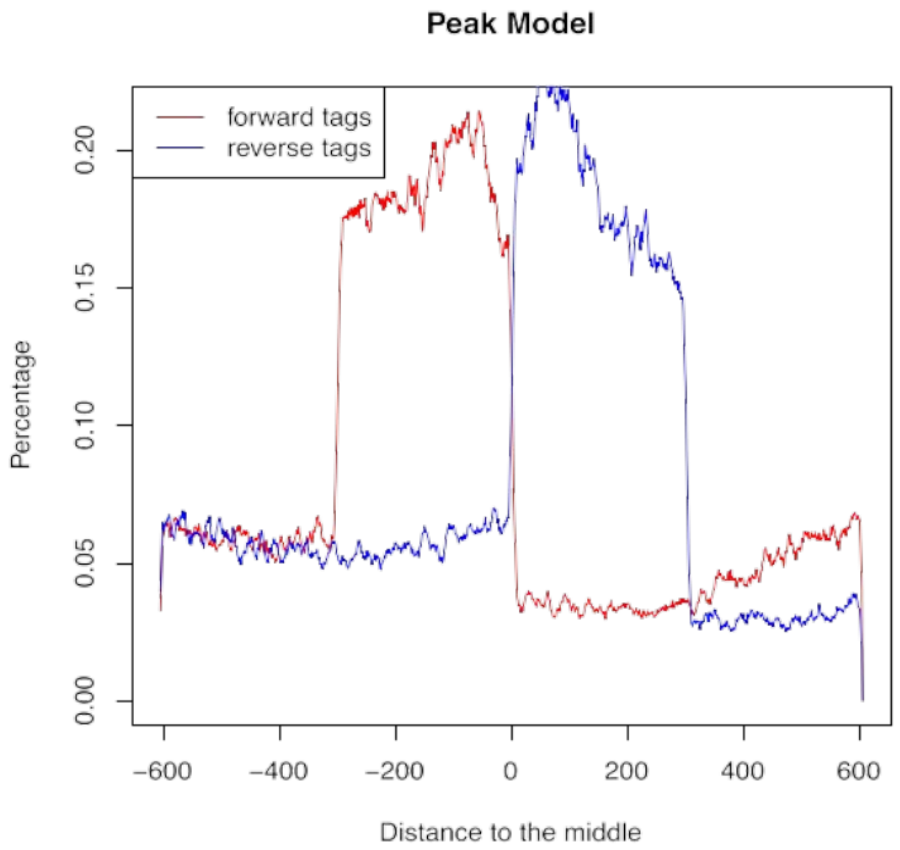
- Most widely used peak caller
- Identifies genome-wide locations of TF binding, histone modification or NFRs (nucleosome-free regions) from ChIP-seq or ATAC-seq data
- Uses controls to eliminate biases due to GC content, mappability, DNA repeats or CNVs
- Can call narrow and broad peaks
- Many settings for optimizing results

# Peak calling with MACS (1)



## Step 1: Estimate fragment length d and adjust read position

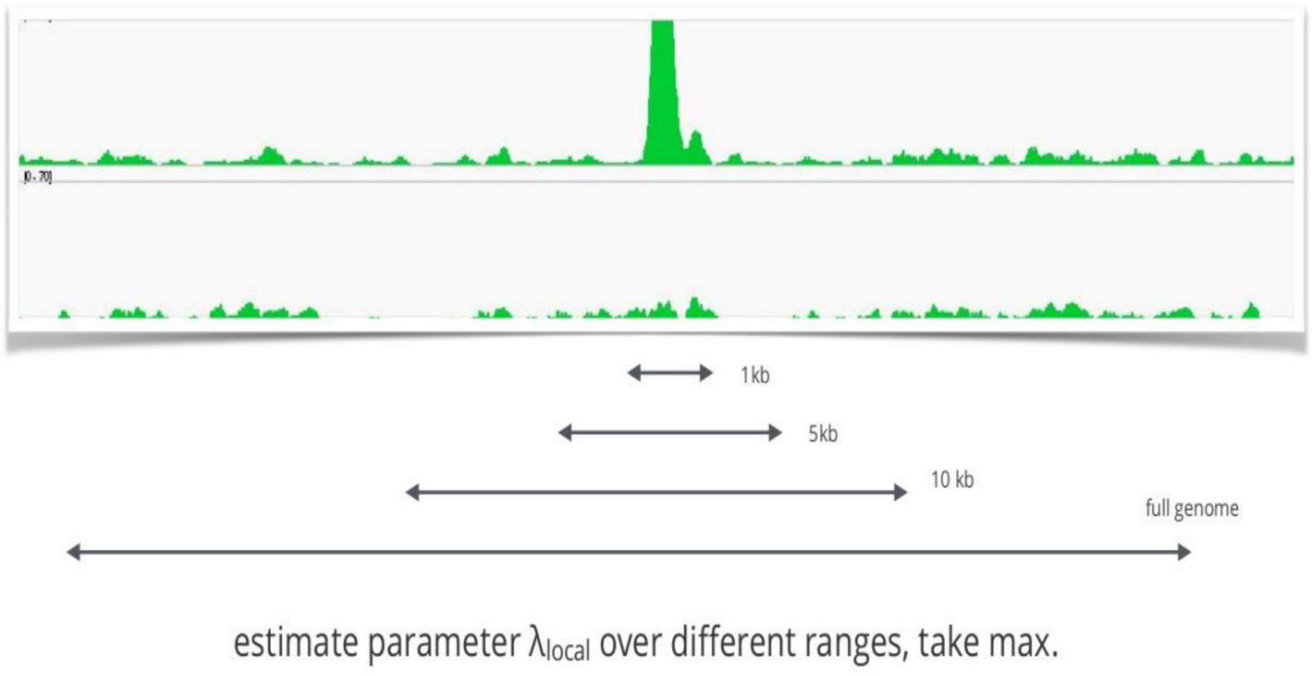
- Slide a window of length 2 * bw bandwidth (half of estimated sonication size) across genome
- Retain windows with (fold-enrichment of treatment / background) > mfold
- Compute the average +/- strand-specific read-densities for these bins



# Peak calling with MACS (2)

## Step 2: Identify local noise

- Slide a window of size 2 * d across treatment and input
- Estimate $\lambda_{local}$ , a parameter of Poisson distribution

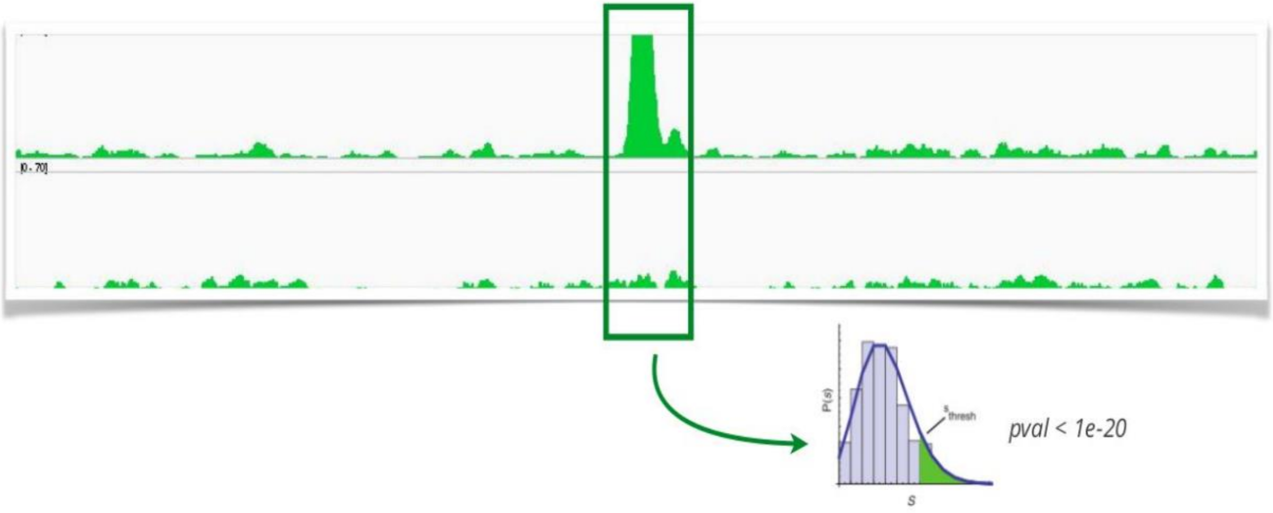


# Peak calling with MACS (3)

上海科技大学 ShanghaiTech University

## Step 3: Identify enriched (peak) regions

- Determine regions with p-value < a cut-off value (e.g. $10^{-20}$)
- Determine summit position within enriched regions as max density




estimate parameter $\lambda_{local}$ over different ranges, take max.



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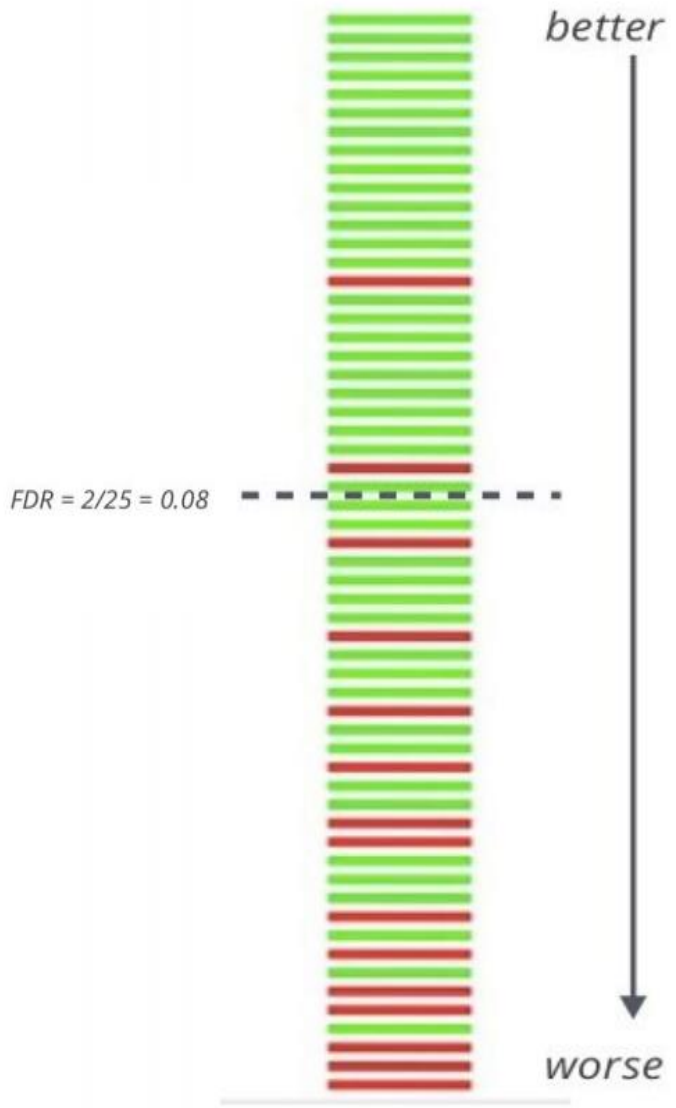
# Peak calling with MACS (4)



## Step 4: Estimate FDR

- Positive peaks (P-values)
- Swap treatment and input, call negative peaks (P-value)

$$\text{FDR} = \frac{\#\ \text{negative peaks with pval} < p}{\#\ \text{positive peaks with pval} < p}$$



# From reads to peaks

- Positive and negative strand reads do not represent true binding sites
- Fragment length d can be detected experimentally or estimated from strand asymmetry in data
- Reads from both strands can be extended to the length of d
- Reads can be shifted towards 3' by d/2

# Appendix B: PEAKS algorithm

# Idea of dynamic programming

- Try to identify the residues one by one from right to left
- Let $f_M(r) = \sum\{h | (x, h) \in M, |x - r| \leq \delta\}$
  - $f_M(r)$ is the sum of all peaks in M whose mass is close to r
- Observation:
  - score(M, $a_1a_2\ldots a_k$) =
    score(M, $a_1a_2\ldots a_{k-1}$) + $f_M(w(a_1a_2\ldots a_k) + 19)$

# Simple dynamic programming solution

- Let $V(r)$ be the maximum score(M, P) among all possible peptide P such that $w(P) = r$
- Our aim is to find $\max_{|r-wt| \leq \delta} V(r)$. Then, by back-tracing, we can recover the peptide
- We have
  - $V(0) = 0$
  - $V(r) = \max_{a \in A} \{ V(r - w(a)) + f_M(r+19) \}$

# Algorithm

**Algorithm Max_Y_Ion**

**Require:** The mass spectrum $M$ and a weight $W$

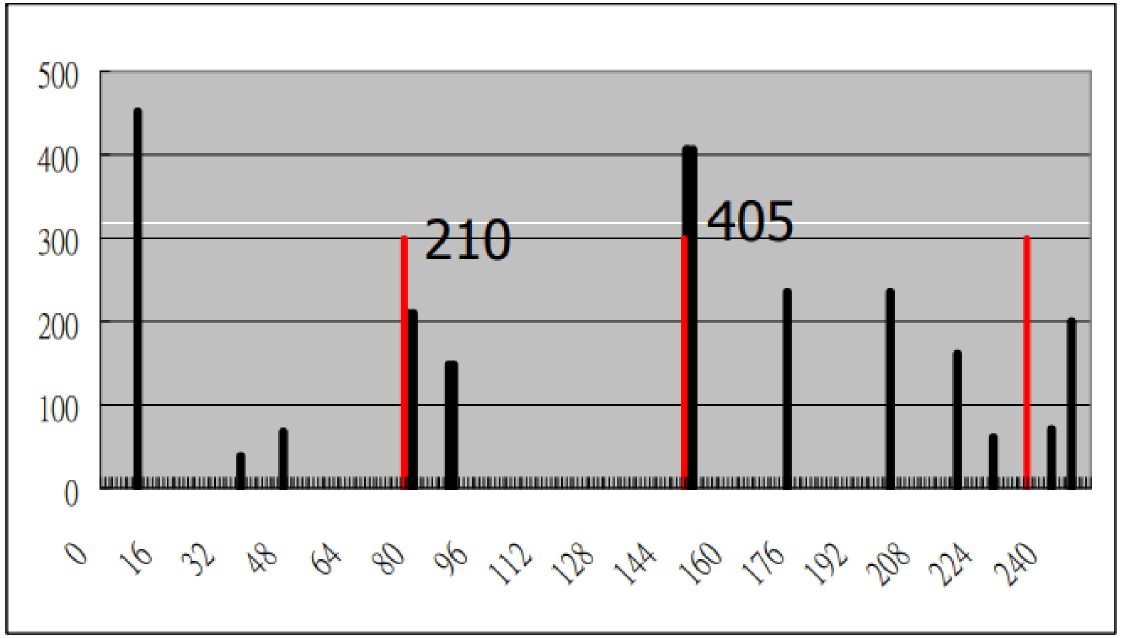
**Ensure:** A peptide $P$ of mass between $W-\delta$ and $W+\delta$ which maximizes $score_Y(M, P)$.

Set $V(r) = 0$ for $r < 0$
**for** $r = 0$ to $W + \delta$ **do**
    $V(r) = \max_{a \in A}\{V(r - w(a)) + f_M(r + 19)\}$
**end for**
$r' = \arg\max_{W-\delta \leq r \leq W+\delta} V(r)$
Through back-tracing, we find the peptide $P$ of mass $r'$ which maximizes $score_Y(M, P)$

# Example

- Given the spectrum M and wt = 215.21
  - V(76.05) = V(0) + 210 = 210 (due to G)
  - V(147.13) = V(76.05) + 405 = 615 (due to A)
  - V(234.21) = V(147.13) + 0 = 615 (due to S)
- By back-tracing, we recover SAG

M



# Time analysis

- We need to fill in the V table with wt entries
- Each entry can be computed in O(|A|) time
- So, the total time complexity is O(|A|∗wt) time

# Better scoring scheme

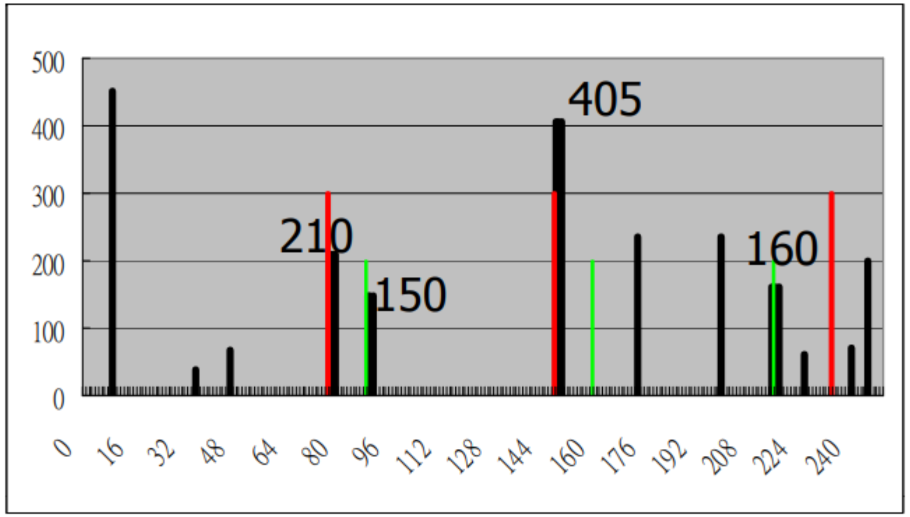
- Besides y-ions, we can also use information from b-ions
- Consider a peptide $P = a_1a_2 \ldots a_k$
  - If M is a spectrum for P, we can find peaks for $m/z = y_i$ or $m/z = b_i$ for $i = 1, 2, \ldots, k$
- So, we redefine the score function score(M, P) as

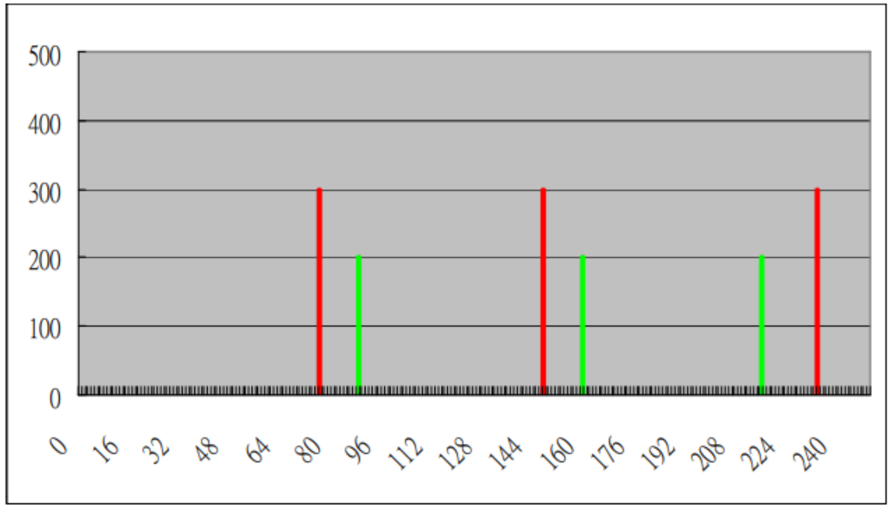
$$\sum \{h \mid (x,h) \in M, |x - y_i| \le \delta \text{ or } |x - b_i| \le \delta \text{ for } i = 1,2,\ldots,k\}$$

# Better scoring scheme example

- E.g. P=SAG
  - $y_1$ = 57.05+71.08+87.08+19 = 234.21
  - $y_2$ = 57.05+71.08+19 = 147.13
  - $y_3$ = 57.05+19 = 76.05
  - $b_1$ = 87.08+1 = 88.08
  - $b_2$ = 87.08+71.08+1 = 159.16
  - $b_3$ = 87.08+71.08+57.05+1 = 216.21

Score(M,P)
= 210+405+150+160
= 925



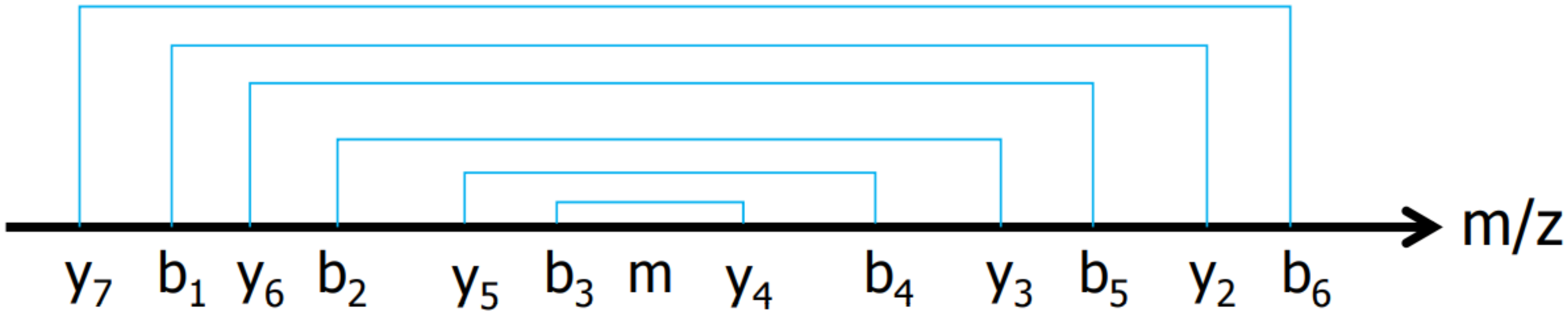

Black peaks: real peaks

Green peaks: artificial b-ions

Red peaks: artificial y-ions

# Observation (I)

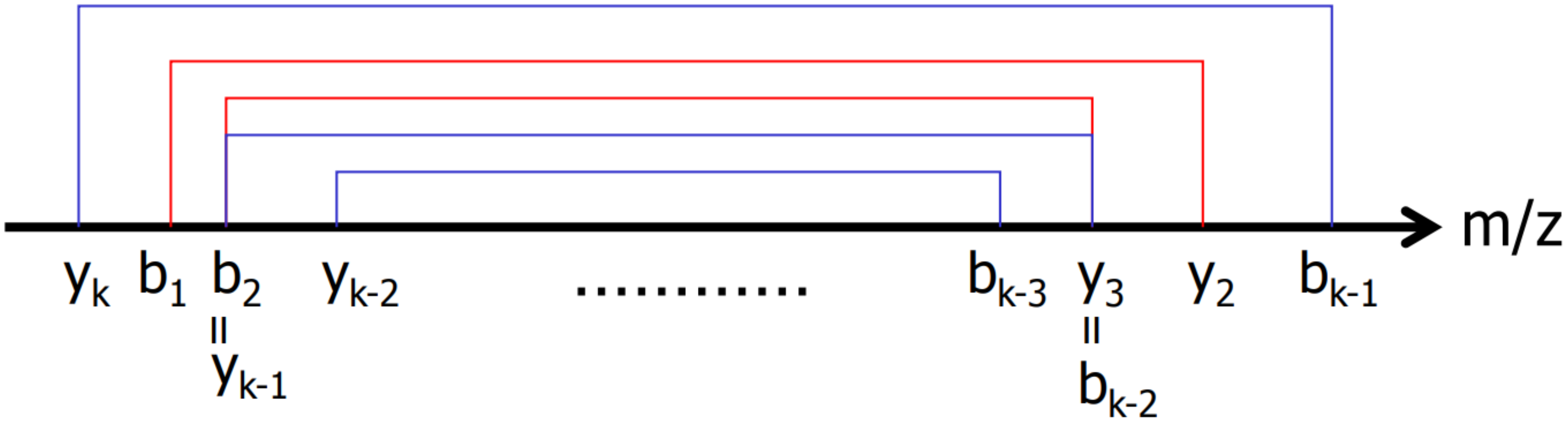
- Suppose $P = a_1 a_2 \ldots a_k$
- $b_i$ is strictly increasing while $y_j$ is strictly decreasing
  - Proof: For any peptide Q and amino acid a, w(Qa), w(aQ) > w(Q)
  - Hence, $b_{i+1} - b_i,\ y_j - y_{j+1} \geq \min_{a \in A} w(a) = 57.05 > 0$
- Note: $b_i + y_{i+1} = w(P) + 20$
  - Hence, $(b_i, y_{i+1})$, for all i=1, 2,…,k, form a set of nested regions
  - For the adjacent nested intervals, the mass difference is at most $\max_{a \in A} w(a) = 186.21$




Consider $P = a_1 a_2 \ldots a_7$.
$m = (w(P)+20)/2$

# Can we solve the problem using previous DP?

- No
- The reason is that, for some masses $y_i$ and $b_j$, their masses may be very close and correspond to the same peak $(x, h) \in M$
- In this case, the previous DP will sum the same peaks twice



# Observation (II)

- Note that the outer-most $L$ intervals are formed by breaking the prefix $a_1 \ldots a_i$ and the suffix $a_j \ldots a_k$, where $i+(k-j+1) = L$
- Let score′ $(M, a_1 \ldots a_i, a_j \ldots a_k)$ be the sum of the intensities of all b-ion and y-ion peaks formed by breaking the peptide P between $a_x$ and $a_{x+1}$ for $x \in \{1,\ldots,i\} \cup \{j-1\ldots,k-1\}$
- Let $f_M(r, s)$ be the sum of all peaks in M which are close to r and wt+20–r but not close to s and wt+20–s (this is used to avoid double counting)
- We have

$$score'(M, a_1 \ldots a_i, a_j \ldots a_k) = \begin{cases} score'(M, a_1 \ldots a_{i-1}, a_j \ldots a_k) + f_M(b_i, y_j) & \text{if } b_i \geq y_j \\ score'(M, a_1 \ldots a_i, a_{j+1} \ldots a_k) + f_M(y_j, b_i) & \text{otherwise} \end{cases}$$

# Solution (a more complicated dynamic programming)

- Let $\hat{a}$ be $\max_{a\in A} w(a)$ = 186.21
- For every $|r - s| \le \hat{a}$, let V(r, s) be the maximum score' $(M,P_1,P_2)$ among all possible $P_1$ and $P_2$, where $w(P_1) = r$ and $w(P_2) = s$

$$V(r,s) = \max \begin{cases} max_{a\in A}\{V(r - w(a), s) + f_M(r+1, s+19)\}, & r \ge s \\ max_{a\in A}\{V(r, s - w(a)) + f_M(s+19, r+1)\}, & r < s \end{cases}$$

$$with\ base\ case\ V(r,s) = 0(r \le 0, s \le 0).$$

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# Solution (a more complicated dynamic programming)

- Aim: Find the best V(r, s) such that wt + 20 = r+s+w(a) for some a ∈ A
- Then, by back-tracking, we can recover the peptide

**Algorithm Sandwich**

**Require:** A mass spectrum $M$, a weight $W$, and an error bound $\delta$

**Ensure:** A peptide $P$ such that $score(M, P)$ is maximized and $|w(P) - W| \leq \delta$

```
Let â = max_{a∈A} w(a)
Initialize all V(r, s) = −∞; Let V(0, 0) = 0
for r = 1 to (W/2 + â) do
    for s = r − â to min{r + â, W − r} do
        for a ∈ A such that r + s + w(a) < W do
            if r < s then
                V(r, s) = max{V(r, s), V(r − w(a), s) + f_M(r + 1, s + 19)}
            else
                V(r, s) = max{V(r, s), V(r, s − w(a)) + f_M(s + 19, r + 1)}
            end if
        end for
    end for
end for
Identify the best V(r, s) among all r, s, a satisfying |r − s| < â and
    |r+s+w(a)−W| < δ. Through back-tracing, we can recover a peptide
    P = P′aP″ where w(P′) = r and w(P″) = s.
```

# Spectrum graph approach

- Another method to recover the peptide is based on spectrum graph, which is defined as follows
- For each mass r in the spectrum M, we generate two vertices of masses r and wt – r
- We also include 2 additional vertices
  - Starting vertex with mass = 0, and
  - Ending vertex with mass = wt

# Generating edges in the spectrum graph



- For every pair of masses r and s, if r – s equals the mass of an amino acid A, we connect x and y with an edge of label A
- Since there may be some missing peaks in the spectrum
  - If r – s equals the total mass of two amino acids $A_1A_2$, we connect x and y with an edge of label $A_1A_2$
  - If r – s equals the total mass of three amino acids $A_1A_2A_3$, we connect x and y with an edge of label $A_1A_2A_3$

# Meaning of a path in the graph

- Every path from start to end corresponds to a possible peptide in the spectrum
- However, there could be many possible paths
- Observe that a vertex has higher probability to be real if all ion types are available
- Hence, we can assign a score depending on whether some ion types are missing
- Then, this is a problem of finding the heaviest path, which can be solved in polynomial time